



AE 541 – Digital Twin in Modern Engineering

Assignment 1: Surrogate Model Development

Rosenbrock (Part A) and Trid (Part B) Functions

September 2026

Student: Bahaa Karawia
Student ID: 202510630
Program: PhD, Aerospace Engineering
Course: Digital Twin in Modern Engineering (AE 541)
Assigned function: Trid function (2 variables)

Abstract. Two benchmark functions are treated as expensive simulations to test how far a cheap surrogate can be trusted. For the Rosenbrock function (common) and the two-variable Trid function (assigned), 60 evaluations from a maximin Latin hypercube design are split into 45 training and 15 held-out test points. Three surrogates are compared: polynomial regression with its degree chosen by five-fold cross-validation on the training set only, a cubic radial basis function (RBF) interpolant, and a Gaussian process (GP). They are assessed with test metrics, side-by-side error maps, learning curves, surrogate-based optimization with verification, and a reduced-box extrapolation test. Cross-validation recovers the true polynomial degrees (4 and 2), and the fitted coefficients match the exact expansions to within 4×10^{-13} . The GP reaches a test normalized root-mean-square error of 7.8×10^{-6} and 1.0×10^{-6} and places the two minima within 0.013 and 6×10^{-4} of the truth. The RBF fails at the domain boundaries and creates a spurious negative minimum in the Rosenbrock valley, which only verification exposes. Outside a reduced training box, errors grow by factors of up to 2300. The hand-derived Trid minimum, $f = -2$ at $(2, 2)$, is confirmed analytically and numerically.

Nomenclature

$\mathbf{A}, \mathbf{P}$	=	RBF kernel matrix and linear-tail matrix
d	=	distance from a surrogate minimum to the known minimum
$f, \hat{f}$	=	true function and surrogate prediction
$\mathcal{F}_k$	=	k -th cross-validation fold
$\mathbf{H}$	=	Hessian matrix
$k(\cdot, \cdot), \mathbf{K}, \mathbf{k}_*$	=	GP kernel, training covariance matrix, and training–test covariances
ℓ_j	=	GP length scale of input j
N, n, n_v	=	main DOE budget, number of training points, number of validation points
n_{terms}	=	number of polynomial coefficients
p	=	total degree of the polynomial surrogate
R^2	=	coefficient of determination
μ_j, s_j	=	training mean and standard deviation of input j (scaling)
$\mathbf{u}$	=	design point in unit-square coordinates
$\mathbf{x}, \mathbf{z}$	=	physical and standardized inputs
$\mathbf{x}^*$	=	location of the global minimum
β	=	polynomial coefficients
ε_{gen}	=	generalization error
$\hat{\mu}, \hat{\sigma}$	=	GP posterior mean and standard deviation
π_j	=	random permutation used by the Latin hypercube
σ_f^2, σ_n^2	=	GP signal and noise variances
$\phi_{\min}$	=	minimum pairwise distance (maximin criterion)
ψ, Ψ	=	vector of monomials and polynomial design matrix
w_i, c_j	=	RBF weights and linear-tail coefficients
CV	=	cross-validation
DOE	=	design of experiments
GP	=	Gaussian process
LHS	=	Latin hypercube sampling
MAE, MaxAE	=	mean and maximum absolute error
(N)RMSE	=	(normalized) root-mean-square error
RBF	=	radial basis function

1 Introduction

High-fidelity analyses such as CFD, finite-element models and coupled aero-structural solvers are too expensive to call thousands of times inside a design loop. The usual answer is a *surrogate model*: a cheap approximation fitted to a small, deliberately chosen set of expensive runs, which then stands in for the analysis during exploration and optimization [1, 2]. A surrogate is useful only if the engineer can say how wrong it is and where. The question behind this study is therefore not “how low can the error go” but “can a cheap model guide a design decision reliably, and inside which bounds” [3].

Two known two-variable functions are treated as if they were expensive simulations. Part A uses the Rosenbrock function [4], common to the whole class; Part B uses the two-variable Trid function [5], assigned to the author. Because both functions are known everywhere, every claim a surrogate makes can be checked against the truth, a luxury that a real project never has. For each function the complete workflow of the course reference notebook [6] is applied: definition of the function and domain, design of experiments (DOE), a leakage-free train/test split, three surrogate types, complexity selection by training-only cross-validation (CV), held-out validation, side-by-side error maps, uncertainty quantification, learning curves, surrogate-based optimization with verification, and an extrapolation test. The shared theory is given once in Sec. 2; results for the two functions follow in Secs. 3 and 4, and Sec. 5 compares them and states where each model can be trusted.

2 Methodology

2.1 Problem Statement

Let the expensive model be a deterministic map $f : \mathcal{X} \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ on the box $\mathcal{X} = [l_1, u_1] \times [l_2, u_2]$. From a design $\mathcal{D} = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^N$ with $y^{(i)} = f(\mathbf{x}^{(i)})$, a surrogate $\hat{f}(\mathbf{x}; \boldsymbol{\theta}) \approx f(\mathbf{x})$ is built. The quantity of interest is the generalization error over the whole domain,

$$\varepsilon_{\text{gen}}^2 = \frac{1}{|\mathcal{X}|} \int_{\mathcal{X}} (f(\mathbf{x}) - \hat{f}(\mathbf{x}))^2 d\mathbf{x}. \quad (1)$$

Equation (1) needs f everywhere, so in practice it can only be *estimated* from data the model has not seen (a held-out test set, or CV). Here it can also be evaluated on a dense 120×120 grid, which is used purely as a diagnostic.

The two functions are listed in Table 1. Both are polynomials: expanding the squares gives

$$f_{\text{R}}(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2 = 100x_1^4 - 200x_1^2x_2 + 100x_2^2 + x_1^2 - 2x_1 + 1, \quad (2)$$

$$f_{\text{T}}(x_1, x_2) = (x_1 - 1)^2 + (x_2 - 1)^2 - x_1x_2 = x_1^2 + x_2^2 - x_1x_2 - 2x_1 - 2x_2 + 2, \quad (3)$$

so the Rosenbrock function is exactly of total degree 4 and the Trid function exactly of degree 2. This fact is known here but is *not* used to choose any model setting; Sec. 3.4 shows that CV recovers it from the data alone.

Table 1: Test functions, domains, known global minima and the exact range of f on each box.

Function	Domain	$\mathbf{x}^*$	$f(\mathbf{x}^*)$	Range of f	Exact degree
Rosenbrock, Eq. (2)	$[-2, 2]^2$	(1, 1)	0	0 to 3609	4
Trid, Eq. (3)	$[-4, 4]^2$	(2, 2)	-2	-2 to 50	2

2.2 Design of Experiments

The main budget is $N = 60$ evaluations, counting training and test points together. All designs are generated in the unit square and mapped linearly to the physical box, so their space-filling quality can be compared independently of the domain size. Three designs are compared at the same N : uniform random sampling; a Latin hypercube (LHS), which splits each axis into N bins and places exactly one point in each bin [7],

$$u_j^{(i)} = \frac{\pi_j(i) - 1 + U_{ij}}{N}, \quad U_{ij} \sim \mathcal{U}(0, 1), \quad (4)$$

where π_j is a random permutation of $\{1, \dots, N\}$; and a *maximin LHS*, the best of 100 independent LHS draws under the maximin criterion [8]

$$\phi_{\min} = \min_{i \neq k} \|\mathbf{u}^{(i)} - \mathbf{u}^{(k)}\|_2. \quad (5)$$

Generating candidate designs costs no function evaluations, so screening them improves the spread for free while keeping the one-dimensional stratification of the LHS. Equations (4) and (5) are written in unit-square coordinates, so $\phi_{\min}$ can be compared between the two boxes. To avoid judging the designs on a single lucky draw, $\phi_{\min}$ is also computed for 500 independent random and LHS realizations.

2.3 Train/Test Split and Data Leakage

The 60 samples are split once, at random (seed 7), into 45 training and 15 test points. The test points are set aside immediately and are used only after every modeling choice has been fixed. *Data leakage* means that information from the test data reaches the fitting process, for example when the input scaling (μ_j, s_j) is computed from all 60 points, or when a degree or kernel bound is chosen by looking at test error; either makes the reported error optimistic. It is avoided here in three ways: (i) the standardization $z_j = (x_j - \mu_j)/s_j$ sits inside a scikit-learn pipeline [9], so every fit, including each CV fold, learns μ_j and s_j from its own training rows only; (ii) the polynomial degree and the GP bound study (Sec. 2.4) are judged by training-only CV; (iii) test and dense-grid errors are reported as diagnostics and never fed back.

2.4 Surrogate Models

Polynomial regression (required). A total-degree- p polynomial in the standardized inputs,

$$\hat{f}(\mathbf{x}) = \sum_{a+b \leq p} \beta_{ab} z_1^a z_2^b = \boldsymbol{\psi}(\mathbf{z})^\top \boldsymbol{\beta}, \quad n_{\text{terms}} = \frac{(p+1)(p+2)}{2}, \quad (6)$$

is fitted by least squares, $\boldsymbol{\beta}^* = \arg \min \|\mathbf{y} - \boldsymbol{\Psi} \boldsymbol{\beta}\|_2^2$, where the rows of the design matrix $\boldsymbol{\Psi}$ are the monomials $\boldsymbol{\psi}(\mathbf{z}^{(i)})^\top$. The coefficients β_{ab} are the only parameters. The system is solved by an SVD-based least-squares routine; when n_{terms} exceeds the number of points, the problem is under-determined and this routine returns the minimum-norm solution, one of infinitely many exact fits. Because an affine input scaling preserves the degree, the fitted model is also an exact degree- p polynomial in the physical $\mathbf{x}$; its physical coefficients are recovered by projecting $\hat{f}$ onto the raw monomials on the grid, which allows a direct comparison with Eqs. (2) and (3).

Cubic radial basis function (RBF) interpolation. The RBF model is a weighted sum of radially symmetric kernels centered on the n training points, plus a linear tail [10]:

$$\hat{f}(\mathbf{z}) = \sum_{i=1}^n w_i \|\mathbf{z} - \mathbf{z}^{(i)}\|^3 + c_0 + c_1 z_1 + c_2 z_2, \quad \sum_i w_i = 0, \quad \sum_i w_i \mathbf{z}^{(i)} = \mathbf{0}. \quad (7)$$

The side conditions in Eq. (7) stop the weights from carrying a linear trend of their own. Imposing $\hat{f}(\mathbf{z}^{(i)}) = y^{(i)}$ gives the symmetric saddle-point system

$$\begin{bmatrix} \mathbf{A} & \mathbf{P} \\ \mathbf{P}^\top & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{w} \\ \mathbf{c} \end{bmatrix} = \begin{bmatrix} \mathbf{y} \\ \mathbf{0} \end{bmatrix}, \quad A_{ik} = \|\mathbf{z}^{(i)} - \mathbf{z}^{(k)}\|^3, \quad \mathbf{P} = [\mathbf{1}, \mathbf{z}_1, \mathbf{z}_2]. \quad (8)$$

The weights w_i and tail coefficients c_0, c_1, c_2 , obtained by solving Eq. (8), are the parameters; the cubic kernel has no shape parameter to tune. The model interpolates the data exactly, which is appropriate for deterministic simulations, but it reproduces exactly only polynomials of degree at most one (the tail).

Gaussian process (GP) regression. The outputs are modeled as a GP with a squared-exponential kernel with automatic relevance determination (ARD) and a small noise term [11]:

$$k(\mathbf{z}, \mathbf{z}') = \sigma_f^2 \exp \left(-\frac{1}{2} \sum_{j=1}^2 \frac{(z_j - z'_j)^2}{\ell_j^2} \right) + \sigma_n^2 \delta_{\mathbf{z}\mathbf{z}'}. \quad (9)$$

In Eq. (9), σ_f^2 is the signal variance (how far the normalized output swings), ℓ_j is the length scale of input j (how far one must move in z_j before the output changes appreciably, so a

small ℓ_j marks a sensitive direction), and σ_n^2 absorbs numerical noise. With $\mathbf{K} = k(\mathbf{Z}, \mathbf{Z})$ and $\mathbf{k}_* = k(\mathbf{Z}, \mathbf{z}_*)$, the posterior mean and variance are

$$\hat{\mu}(\mathbf{z}_*) = \mathbf{k}_*^\top \mathbf{K}^{-1} \mathbf{y}, \quad \hat{\sigma}^2(\mathbf{z}_*) = k(\mathbf{z}_*, \mathbf{z}_*) - \mathbf{k}_*^\top \mathbf{K}^{-1} \mathbf{k}_*. \quad (10)$$

The hyperparameters $\boldsymbol{\theta} = (\sigma_f, \ell_1, \ell_2, \sigma_n)$ maximize the log marginal likelihood

$$\log p(\mathbf{y} \mid \mathbf{Z}, \boldsymbol{\theta}) = -\frac{1}{2} \mathbf{y}^\top \mathbf{K}^{-1} \mathbf{y} - \frac{1}{2} \log |\mathbf{K}| - \frac{n}{2} \log 2\pi. \quad (11)$$

Equation (11) is maximized with normalized outputs, eight random restarts (nine optimizer runs in total), and the bounds $\sigma_f^2 \in [10^{-3}, 10^4]$, $\ell_j \in [10^{-2}, 10^3]$ and $\sigma_n^2 \in [10^{-12}, 10^{-2}]$; the initial values and all other settings are listed in the configuration block of the script. The GP was added as a third model because it provides the uncertainty estimate $\hat{\sigma}$ needed for step 9 of the assignment, and because it is the standard surrogate in aerospace design optimization [2, 12]. Whenever a hyperparameter ends at a bound, the fit is repeated with $\sigma_f^2 \leq 10^6$ and $\leq 10^8$ and the alternatives are compared by training CV only. Optimizer runs that end with an abnormal line-search termination are counted and reported.

2.5 Complexity Selection

The polynomial degree p of Eq. (6) is swept over $p = 1, \dots, 8$ and scored by the pooled five-fold CV RMSE on the 45 training points,

$$\text{RMSE}_{\text{CV}} = \sqrt{\frac{1}{n} \sum_{k=1}^5 \sum_{i \in \mathcal{F}_k} (y^{(i)} - \hat{f}^{(-k)}(\mathbf{x}^{(i)}))^2}, \quad (12)$$

where $\hat{f}^{(-k)}$ is refitted, scaler included, without fold $\mathcal{F}_k$ [13]. The choice of p is a bias–variance trade-off [13]: too low a degree leaves a systematic (bias) error that more data cannot remove, whereas too high a degree makes the fit sensitive to the particular sample (variance) and, numerically, to the conditioning of Ψ . Each CV training fold holds 36 points, so degree 7 ($n_{\text{terms}} = 36$) is exactly determined and degree 8 ($n_{\text{terms}} = 45$) is under-determined inside the folds. With noiseless polynomial data every degree at or above the true one gives an error at round-off level, and the arg-min among those values is decided by floating-point noise. The selection rule is therefore a parsimony rule: choose the smallest p whose CV RMSE is within 5% of the best value, or below $10^{-9} \text{std}(y_{\text{train}})$, which is treated as exact. Only the CV error of Eq. (12) enters the choice; the test and dense-grid errors are computed after the degree has been locked and are plotted only as diagnostics.

2.6 Validation Metrics

For n_v validation points with truth y_i and prediction $\hat{y}_i$,

$$\text{RMSE} = \sqrt{\frac{1}{n_v} \sum_i (y_i - \hat{y}_i)^2}, \quad \text{MAE} = \frac{1}{n_v} \sum_i |y_i - \hat{y}_i|, \quad R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \quad (13)$$

$$\text{MaxAE} = \max_i |y_i - \hat{y}_i|, \quad \text{NRMSE} = \frac{\text{RMSE}}{\max(y_{\text{test}}) - \min(y_{\text{test}})}. \quad (14)$$

The metrics of Eqs. (13) and (14) are computed on the test set; on the dense grid, NRMSE is normalized by the range of the grid values instead. Every model is compared with the trivial baseline $\hat{y} = \text{mean}(y_{\text{train}})$; a surrogate that does not beat it has learned nothing. RMSE and MAE carry the units of f , NRMSE makes the two functions comparable, and MaxAE is the worst-case number that usually matters in design. CV is used to select, the test set gives the final independent check, and the dense grid shows what both estimates miss.

2.7 Uncertainty, Sample Size, Optimization and Extrapolation

Uncertainty. The GP standard deviation $\hat{\sigma}$ of Eq. (10) is mapped over the grid and compared with the true $|f - \hat{f}|$ through the Pearson and Spearman correlations and the fraction of grid points with $|f - \hat{f}| \leq 2\hat{\sigma}$. The small diagonal jitter $\alpha = 10^{-10}$ that scikit-learn adds to $\mathbf{K}$ for numerical stability, together with σ_n^2 , puts a floor of about $\sqrt{\alpha + \sigma_n^2} \text{std}(y_{\text{train}})$ under $\hat{\sigma}$; this floor is reported with the results.

Learning curve. Each model is refitted on fresh LHS designs of $N = 20, 30, 45, 65, 90$ training points, five independent draws per size, and scored by RMSE on the full grid and on a central box whose sides are 75% of the domain sides (56% of its area). The core score separates edge effects from interior accuracy. The polynomial keeps its CV-selected degree at every N because $n_{\text{terms}} \leq 20$ in both cases; no degree reduction was needed.

Optimization. Each surrogate is minimized by bounded Nelder–Mead [14] from 20 LHS start points. Nelder–Mead is derivative free, which avoids the finite-difference failure on GP predictions noted in the reference notebook [6]. End points closer than 2% of the box width (0.08 for Rosenbrock, 0.16 for Trid) are merged into one candidate. Each distinct candidate is then evaluated once with the true function, and the predicted value, the true value and the distance to the known minimum are tabulated. The known minimum is never inserted as a result. The same multi-start search on the true function confirms the reference minimum numerically.

Extrapolation. Each model is retrained on 45 LHS points inside a reduced box (the central quarter of the domain by area) and evaluated over the full grid; errors inside and outside the training box are reported separately.

2.8 Reproducibility

All results come from the single script `202510630_Bahaa_Karawia.py` (Appendix A), run with Python 3.11.9, NumPy 2.4.6 [15], SciPy 1.17.1 [16], scikit-learn 1.9.1 [9], Matplotlib 3.11.2 [17] and pandas 3.0.6 [18]. The master seed is 7, and every random draw derives from it through fixed offsets set in the configuration block. The command `python 202510630_Bahaa_Karawia.py` writes every figure and table of this report to `./outputs` in about two minutes. The same unit-square design is used for both functions, so differences between Parts A and B come from the functions, not from the sampling. The workflow and parts of the code (LHS, RBF system, GP set-up) are adapted from the course notebook [6].

3 Results, Part A: Rosenbrock Function

3.1 Function, Domain and Minimum

Figure 1 shows the Rosenbrock function on $[-2, 2]^2$. The minimum $f = 0$ at $(1, 1)$ lies at the end of the curved “banana” valley $x_2 = x_1^2$; the script checks $f(1, 1) = 0$ before any fitting. Across the valley the function rises steeply, reaching 3609 at the corner $(-2, -2)$, while along its floor it reduces to $(1 - x_1)^2$ and varies only between 0 and 5.8. The interesting region (small f) is a thin curved band that covers a tiny fraction of the domain, so any global error metric will be dominated by the large corner values.

3.2 Design of Experiments

The three designs of Sec. 2.2 are shown in Fig. 2, and their maximin distances are listed in Table 2. The single random design leaves visible holes and clusters and its axis projections (red ticks) pile up; the LHS projections are evenly spaced by construction. Over 500 realizations, the median $\phi_{\min}$ of LHS (0.0217) is almost twice that of random sampling (0.0115), and the

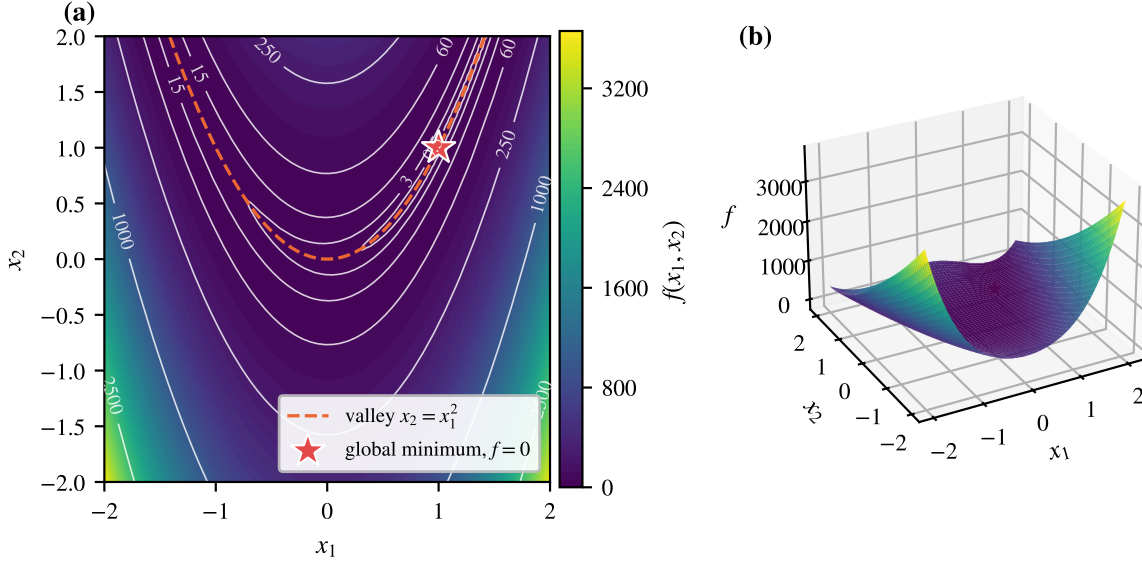


Figure 1: Rosenbrock function on $[-2, 2]^2$. (a) Filled contour map with iso-lines at $f = 0.5, 3, 15, 60, 250, 1000, 2500$; the dashed line is the valley floor $x_2 = x_1^2$ and the star marks the global minimum $f = 0$ at $(1, 1)$. (b) Three-dimensional surface of the same field, with the minimum marked.

5th percentile is three times larger (0.0105 against 0.0035), so the LHS rarely produces the near-duplicate points that waste a random design. The maximin LHS used here reaches $\phi_{\min} = 0.0452$, higher than every one of the 500 plain LHS draws (maximum 0.0432) and almost twice the random design at the same cost.

The budget $N = 60$ was chosen from the models to be fitted. With 45 training points a degree-4 polynomial (15 coefficients) has three points per coefficient on the full training set and 2.4 per coefficient inside a CV fold, above the usual rule of thumb $n \geq 2n_{\text{terms}}$; the learning curve of Sec. 3.9 then confirms that the GP error falls by two orders of magnitude up to $N = 45$ and only by a factor of three between 45 and 90, so a larger budget would buy comparatively little.

Table 2: Space-filling quality of the candidate designs, measured by the maximin distance $\phi_{\min}$ in the unit square ($N = 60$, higher is better).

Design	$\phi_{\min}$ (seed 7)	Median over 500 draws	5th percentile over 500 draws
Uniform random	0.0229	0.0115	0.0035
Latin hypercube	0.0251	0.0217	0.0105
Maximin LHS (used)	0.0452	—	—

3.3 Train/Test Split

Figure 3 shows the 45 training and 15 test points. The test points are spread over the whole box. Three lie on the valley floor, at $(-0.83, 0.68)$, $(-0.46, 0.22)$ and $(-0.10, 0.06)$, where $f < 3.4$. Two, $(0.91, 1.36)$ and $(1.45, 0.88)$, lie within 0.5 of the minimum but on the steep flanks ($f = 27$ and 151), not in the low- f valley near it, and one, $(-1.47, -1.85)$, approaches the corner $(-2, -2)$. Fifteen points cannot sample every feature, which is one reason CV and the dense-grid diagnostic are reported alongside the test metrics.

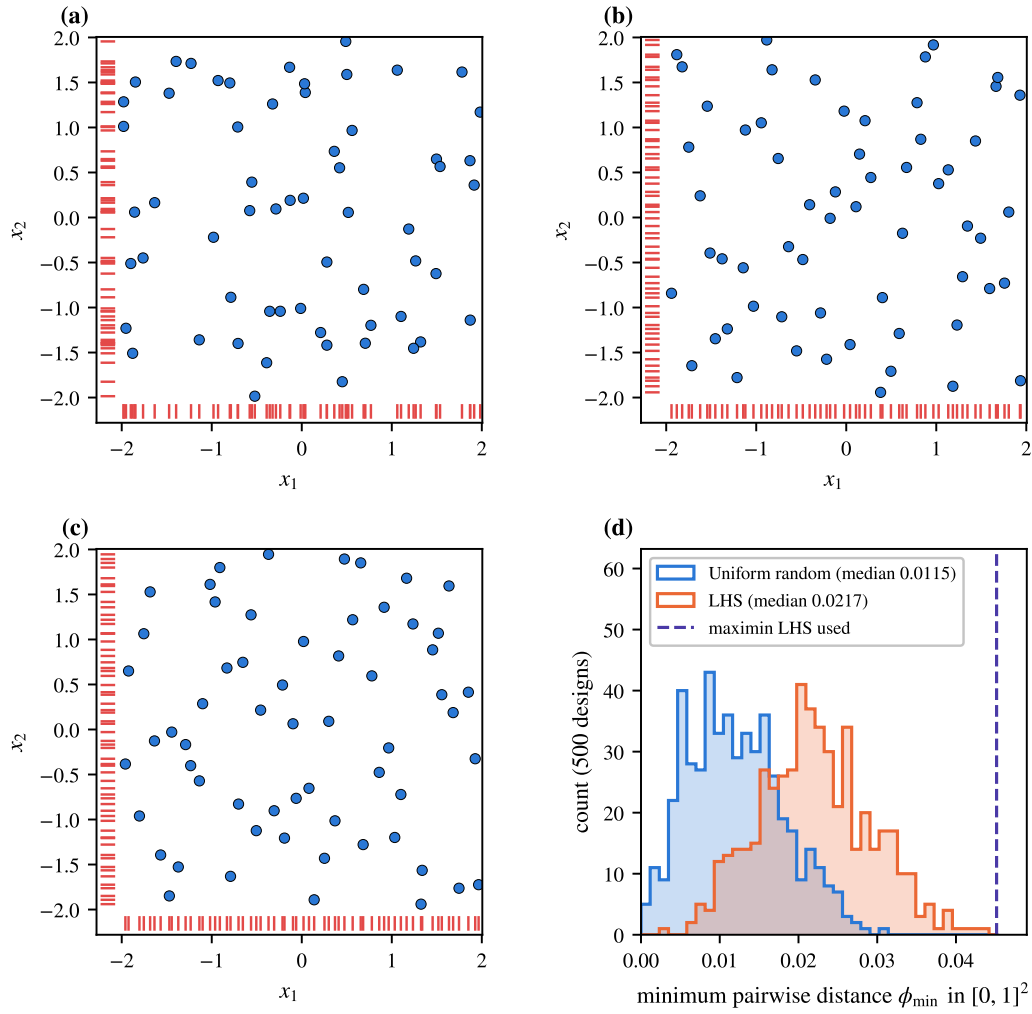


Figure 2: Comparison of designs with the same budget $N = 60$, shown in the Rosenbrock box. (a) Uniform random, $\phi_{\min} = 0.0229$. (b) Latin hypercube, $\phi_{\min} = 0.0251$. (c) Maximin LHS (best of 100 LHS draws), $\phi_{\min} = 0.0452$, used for all results. Red ticks along the left and bottom edges are the one-dimensional projections of the points. (d) Distribution of $\phi_{\min}$, measured in the unit square, over 500 independent random and LHS designs; the dashed line is the maximin LHS actually used. The Trid study uses the same unit-square design scaled to $[-4, 4]^2$.

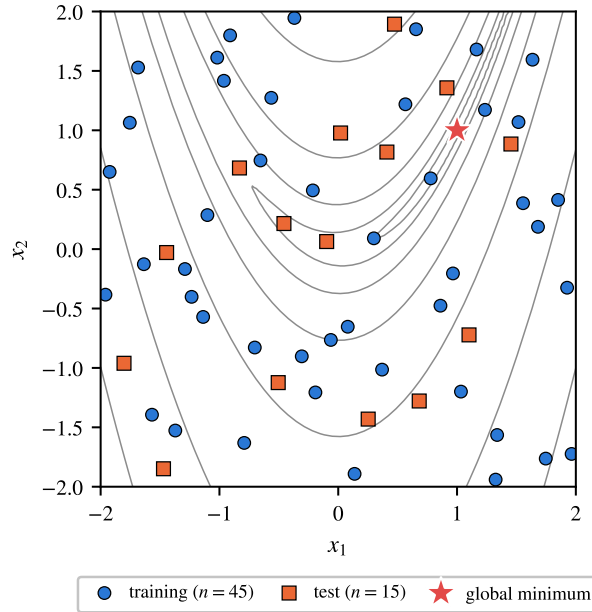


Figure 3: Rosenbrock: locations of the 45 training points and the 15 held-out test points over iso-lines of f ; the star is the global minimum.

3.4 Polynomial Degree Selection

The degree sweep is shown in Fig. 4 and listed in Table 3. For $p \leq 3$ the train, CV and test errors are all large (111 to 607) and close to one another: the model is too rigid and the error is bias, which more data would not remove. At $p = 4$ every curve collapses by about 14 orders of magnitude to round-off ($\text{RMSE}_{\text{CV}} = 1.27 \times 10^{-12}$), because the hypothesis space now contains the truth. The plain CV arg-min would have picked $p = 5$, whose CV RMSE is lower by only 1.4×10^{-13} , a floating-point difference; the parsimony rule selects $p = 4$, the true degree, from the data alone. The recovered physical coefficients match Eq. (2) to within 3.8×10^{-13} , and every coefficient that should vanish is below 2.5×10^{-13} .

Beyond $p = 4$ no classic over-fitting appears in the *train* or *test* error: the data are noiseless and the truth is inside every larger space, so the extra coefficients are simply fitted as zero. What does appear is numerical. The condition number of the standardized design matrix grows from 33 at $p = 4$ to 2.3×10^3 at $p = 7$ and 2.1×10^5 at $p = 8$, and the CV error climbs to 2.6×10^{-11} at $p = 7$ (36 terms on 36 fold points) and jumps to 82.9 at $p = 8$. At $p = 8$ each fold has 45 unknowns and 36 equations: the minimum-norm solution passes exactly through the fold's points but is not the true polynomial, and it is badly wrong on the left-out fold. On the full training set the same degree has a square 45×45 system whose unique solution is the truth again (test RMSE 1.9×10^{-10}). The gap between the CV curve and the train curve is therefore a signal of under-determination, not of noise-driven over-fitting.

3.5 Held-Out Validation and Cross-Validation

Table 4 lists the test metrics and Table 5 the CV and dense-grid errors; the parity plots are in Fig. 5. All three models beat the mean baseline, whose R^2 is even negative (-0.069) because the test mean differs from the training mean. The polynomial is exact to round-off. The GP reaches a test RMSE of 0.0138 (NRMSE 7.8×10^{-6}) with a worst test error of 0.047. The cubic RBF has a test RMSE of 33.3 and a MaxAE of 116, but its $R^2 = 0.9962$ still looks excellent: the denominator of R^2 is the huge variance of the corner values, which is exactly why R^2 alone

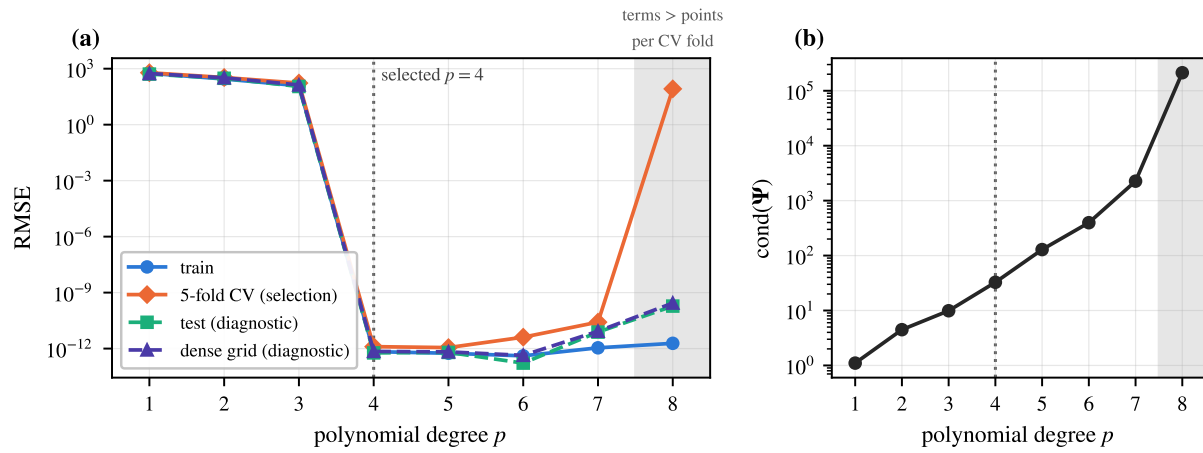


Figure 4: Rosenbrock: polynomial complexity study. (a) RMSE against degree p : training error, five-fold CV error on the training set (the only curve used for selection), and the held-out test and dense-grid errors added afterwards as diagnostics; the dotted line marks the selected $p = 4$ and the shaded band marks degrees with more coefficients than points in a CV training fold. (b) Condition number of the standardized monomial design matrix.

Table 3: Polynomial degree sweep on the 45 training points: number of coefficients, condition number of the standardized design matrix, and five-fold CV RMSE for both functions. Selected degrees in bold.

p	n_{terms}	Points per fold	$\text{cond}(\Psi)$	CV RMSE, Rosenbrock	CV RMSE, Trid
1	3	36	1.10	6.07×10^2	1.01×10^1
2	6	36	4.48	3.36×10^2	1.04×10^{-14}
3	10	36	9.90	1.70×10^2	2.89×10^{-14}
4	15	36	32.6	1.27×10^{-12}	1.82×10^{-14}
5	21	36	129	1.13×10^{-12}	2.58×10^{-14}
6	28	36	396	4.09×10^{-12}	3.37×10^{-14}
7	36	36	2.26×10^3	2.61×10^{-11}	5.56×10^{-12}
8	45	36	2.13×10^5	8.29×10^1	9.17×10^{-1}

flatters a model on a function with this dynamic range.

The three error estimates in Table 5 do not agree, and the disagreement is informative. For the RBF the CV RMSE (78.2) is more than twice the test RMSE (33.3), and the dense grid (66.1) sits much closer to CV. The 15 test points happen to miss the worst errors, which lie in the corner itself and on the flanks, while CV predicts all 45 training points from models built on 36, which is more pessimistic but closer to the truth here. A single small test set can make a surrogate look better than it is, which is why CV and the error maps are needed alongside it.

Table 4: Held-out test metrics (15 points never used in fitting or selection). The range of y_{test} used for NRMSE is 1780.9 for Rosenbrock and 28.07 for Trid.

Function	Model	RMSE	MAE	R^2	MaxAE	NRMSE
Rosenbrock	Baseline, $\text{mean}(y_{\text{train}})$	558.4	458.8	-0.069	1270.2	0.314
	Polynomial, $p = 4$	5.87×10^{-13}	5.22×10^{-13}	1.000000	8.98×10^{-13}	3.30×10^{-16}
	Cubic RBF	33.27	18.21	0.996204	115.55	1.87×10^{-2}
	GP	1.38×10^{-2}	7.48×10^{-3}	1.000000	4.70×10^{-2}	7.75×10^{-6}
Trid	Baseline, $\text{mean}(y_{\text{train}})$	9.725	8.375	-0.262	15.12	0.347
	Polynomial, $p = 2$	7.91×10^{-15}	6.90×10^{-15}	1.000000	1.55×10^{-14}	2.82×10^{-16}
	Cubic RBF	4.25×10^{-2}	2.04×10^{-2}	0.999976	0.155	1.51×10^{-3}
	GP	2.91×10^{-5}	1.87×10^{-5}	1.000000	8.98×10^{-5}	1.04×10^{-6}

Table 5: Five-fold CV RMSE on the training partition (used for selection), held-out test RMSE (final check), and dense-grid diagnostics over 14 400 points (available only because the truth is known). Grid NRMSE is normalized by the range of f on the grid.

Function	Model	CV RMSE	Test RMSE	Grid RMSE	Grid MaxAE	Grid NRMSE
Rosenbrock	Polynomial, $p = 4$	1.27×10^{-12}	5.87×10^{-13}	7.06×10^{-13}	3.13×10^{-12}	1.96×10^{-16}
	Cubic RBF	78.24	33.27	66.12	917.8	1.83×10^{-2}
	GP	5.78×10^{-2}	1.38×10^{-2}	3.27×10^{-2}	0.699	9.06×10^{-6}
Trid	Polynomial, $p = 2$	1.04×10^{-14}	7.91×10^{-15}	8.14×10^{-15}	2.84×10^{-14}	1.57×10^{-16}
	Cubic RBF	0.178	4.25×10^{-2}	0.156	2.72	2.99×10^{-3}
	GP	4.69×10^{-5}	2.91×10^{-5}	5.58×10^{-5}	9.57×10^{-4}	1.07×10^{-6}

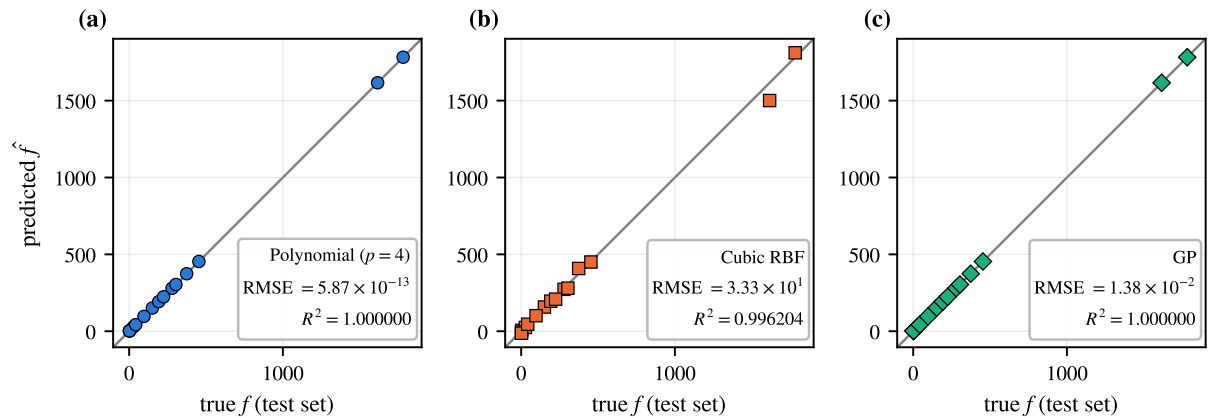


Figure 5: Rosenbrock: parity plots of predicted against true f on the 15 held-out test points for (a) the degree-4 polynomial, (b) the cubic RBF and (c) the GP. The solid line is perfect prediction; the inset gives the test RMSE and R^2 .

3.6 GP Hyperparameters and the Banana Valley

The fitted length scales are $\ell_1 = 2.57$ and $\ell_2 = 23.4$ in standardized units, or 3.11 and 27.4 in physical units. The ratio $\ell_2/\ell_1 \approx 8.8$ says that the function is roughly an order of magnitude more sensitive to x_1 than to x_2 , which agrees with Eq. (2): x_1 enters through the quartic $100x_1^4$ and the coupling $-200x_1^2x_2$, whereas x_2 enters only quadratically. The ARD kernel, however, measures sensitivity only along the coordinate axes. The valley is curved: inside the box it exists for $|x_1| \leq \sqrt{2}$, and its tangent direction $(1, 2x_1)$ rotates from $(1, -2.83)$ to $(1, 2.83)$ along it, so no pair of axis-aligned length scales can describe “slow along the valley, fast across it”. The two fitted values are a compromise averaged over the whole box, not a description of the valley.

The signal variance ended at its upper bound $\sigma_f^2 = 10^4$ and the noise variance close to its lower bound (1.2×10^{-12} against 10^{-12}), and 3 of the 9 optimizer runs ended with an abnormal line-search termination; the run with the highest likelihood is kept. A stationary squared-exponential kernel approaches a polynomial only in the limit of large amplitude and long length scales [11], so the likelihood keeps pushing σ_f^2 and ℓ_j upward, into a flat and poorly conditioned region. Table 6 shows that widening the bound does not settle the matter. With $\sigma_f^2 \leq 10^6$ the likelihood rises to 214.7, but the training CV error gets worse (0.079 against 0.058). With $\sigma_f^2 \leq 10^8$ the likelihood falls to 183.6, which cannot be the global optimum, because the 10^4 solution is still admissible: the optimizer has found a different local optimum. The marginal likelihood is therefore multimodal and unreliable here, and the choice between the fits was made by training CV, which favors the 10^4 bound. The bound acts as a mild regularizer. The principled remedy would be universal Kriging, a GP around a polynomial trend [2], which is beyond the scope of this assignment.

Table 6: GP hyperparameters refitted with wider bounds on the normalized signal variance σ_f^2 (physical length scales). The alternatives are judged by training CV only; the first row of each block is the model used, for which 3 (Rosenbrock) and 8 (Trid) of the 9 optimizer runs ended with an abnormal line-search termination. A wider bound still admits the narrower optimum, so a lower likelihood indicates a different local optimum, not a worse model class.

Function	Bound on σ_f^2	Fitted σ_f^2	ℓ_1	ℓ_2	σ_n^2	$\log p(\mathbf{y} \mid \boldsymbol{\theta})$	CV RMSE
Rosenbrock	10^4 (used)	10^4 (at bound)	3.11	27.4	1.2×10^{-12}	197.8	5.78×10^{-2}
	10^6	10^6 (at bound)	4.80	99.8	9.8×10^{-10}	214.7	7.89×10^{-2}
	10^8	3.77×10^6	6.92	126.4	4.8×10^{-8}	183.6	6.10×10^{-2}
Trid	10^4 (used)	10^4 (at bound)	37.5	37.2	1.9×10^{-12}	302.6	4.69×10^{-5}
	10^6	10^6 (at bound)	125.6	140.9	9.6×10^{-9}	260.1	2.74×10^{-4}
	10^8	1.14×10^5	48.8	51.8	6.7×10^{-9}	254.6	9.34×10^{-5}

3.7 Side-by-Side Visual Comparison

Figures 6 to 8 give the mandatory side-by-side comparison for the CV-best surrogate (the polynomial) and for the other two models; Fig. 9 gives the 3-D pair for the best surrogate. The polynomial error field (right panel of Fig. 6) is pure round-off, at most 3.1×10^{-12} against a signal of 3609. Its smooth large-scale pattern reflects floating-point cancellation in the quartic terms, not model error. Because this map carries no modeling information, the GP and RBF maps are where the failure patterns can be read. On every map of f and $\hat{f}$ in this report, the color scale extends slightly below the minimum of f on the grid (by 0.1% of the range of f , i.e. to -3.6 for Rosenbrock and -2.05 for Trid), and surrogate values below that limit are drawn in red, so a clearly impossible prediction cannot hide in the darkest color.

For the GP (right panel of Fig. 7) the error is below 0.05 almost everywhere and concentrates in the corner $(-2, -2)$, where it reaches 0.70. That corner combines the steepest part of the surface ($f = 3609$) with the largest corner gap in the design: its nearest training point is 0.74 away, against 0.28 to 0.57 for the other three corners, and no point lies beyond the boundary to constrain the fit. For the RBF (right panel of Fig. 8) the error is three orders of magnitude larger and has the same hot spot, reaching 918 in the corner $(-2, -2)$. It also shows negative lobes of several hundred on both steep flanks near $(\pm 1.7, -0.9)$, where the interpolant overshoots between samples, and the surrogate panel shows closed contours inside the valley that do not exist in the truth, enclosing a red region where $\hat{f} < -3.6$ and falls to -16.8 , although $f \geq 0$ everywhere. A cubic spline with a linear tail cannot reproduce a quartic, and near the boundary it relaxes toward its linear tail, whereas the true surface curves up fastest precisely at the edges [19]. In all maps the largest errors sit in gaps between samples or outside the convex hull of the training points, never at the samples themselves.

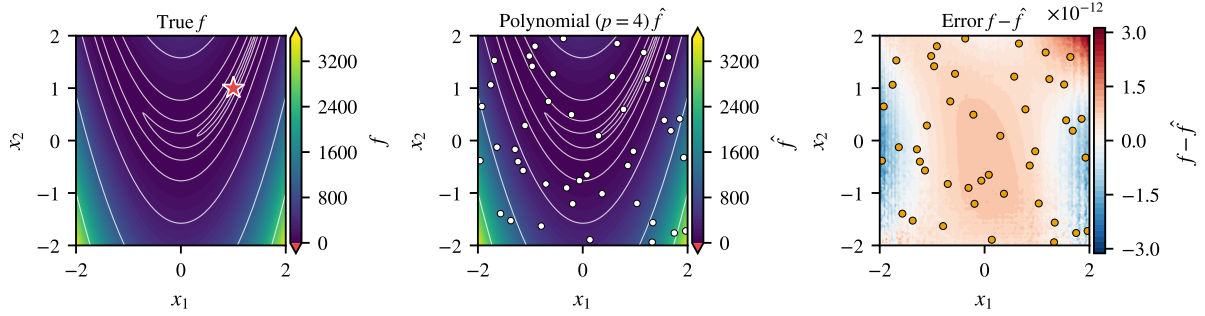


Figure 6: Rosenbrock, best surrogate by CV (degree-4 polynomial). Left: true f on the 120×120 grid with the minimum marked. Center: surrogate $\hat{f}$ on the same grid with identical color limits, training points overlaid. Right: error $f - \hat{f}$ on a diverging scale centered at zero with training points overlaid; the maximum $|e|$ is 3.1×10^{-12} (round-off). White iso-lines are drawn at the same levels in the first two panels.

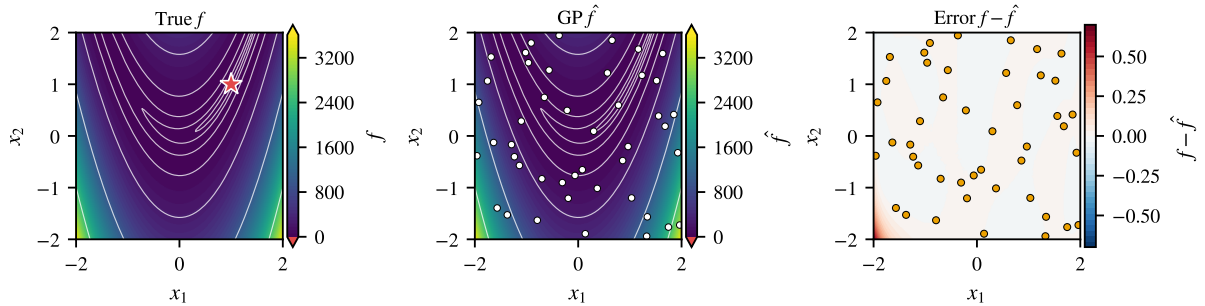


Figure 7: Rosenbrock, GP surrogate: true f , surrogate $\hat{f}$ with identical color limits, and error $f - \hat{f}$ (diverging scale centered at zero, maximum $|e| = 0.699$ in the corner $(-2, -2)$), with training points overlaid on the surrogate and error panels.

3.8 Uncertainty Quantification

Figure 10 compares the GP standard deviation $\hat{\sigma}$, which is available without knowing the truth, with the actual error $|f - \hat{f}|$, which is not. The two maps share their structure: both are smallest near the samples and largest in the corners, especially $(-2, -2)$. Over the grid, the Pearson correlation is 0.70 and the Spearman rank correlation 0.67, and 97.8% of grid points satisfy $|f - \hat{f}| \leq 2\hat{\sigma}$. The 2.2% outside the band matter most: they lie on the left side of the

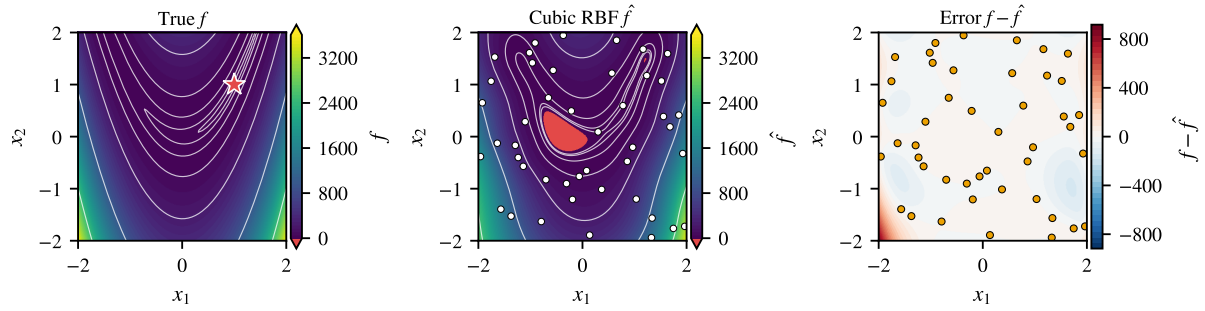


Figure 8: Rosenbrock, cubic RBF surrogate: true f , surrogate $\hat{f}$ with identical color limits, and error $f - \hat{f}$ (diverging scale centered at zero, maximum $|e| = 918$ in the corner $(-2, -2)$), with training points overlaid. Negative lobes on both steep flanks mark overshoot between samples. The red regions in the center panel are where the surrogate predicts values more than 0.1% of the range of f below its minimum ($\hat{f} < -3.6$): a large one reaching -16.8 inside closed contours that the truth does not have, and a small one near $(1.2, 1.45)$ in the upper arm of the valley.

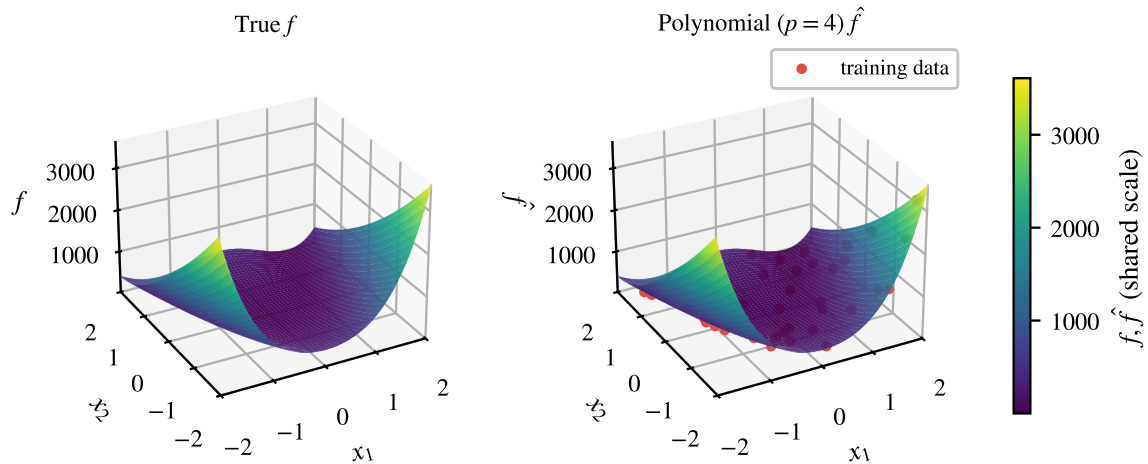


Figure 9: Rosenbrock: 3-D surfaces of the true function and of the degree-4 polynomial surrogate with identical axis limits, color limits and viewing angle; red dots are the training data.

domain (centroid $(-1.50, -0.61)$) and include the worst error of all, 0.70, which exceeds twice the largest predicted standard deviation ($2\hat{\sigma}_{\max} = 0.50$). The band therefore fails exactly where the error is largest. Away from the corners, $\hat{\sigma}$ is close to its numerical floor of 6.8×10^{-3} (median 5.9×10^{-3} in the central box, the same order as the value 7.0×10^{-3} at the training points), so the useful information in $\hat{\sigma}$ lies in its rise toward the edges, up to 0.25. The two quantities are related because both grow with distance from the data. They are not the same quantity: $\hat{\sigma}$ depends only on the sample locations and the fitted kernel, not on the observed values at the prediction point, so it measures how poorly constrained the model is under its own assumptions, not how wrong it actually is. Panel (c) shows this directly: at a given $\hat{\sigma}$ the true error spreads over two to three decades, and a lower branch of points has errors ten times smaller than $\hat{\sigma}$. The GP is a useful *guide* to where to sample next, but not a certificate of accuracy.

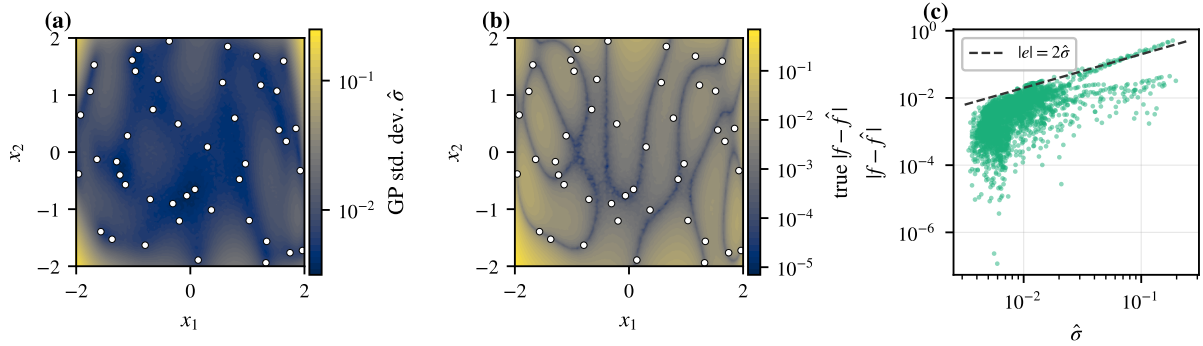


Figure 10: Rosenbrock, GP uncertainty. (a) Predicted standard deviation $\hat{\sigma}$ (log color scale), computable without the truth. (b) True absolute error $|f - \hat{f}|$ (log color scale). (c) $|f - \hat{f}|$ against $\hat{\sigma}$ for 3000 random grid points on logarithmic axes; the dashed line is $|e| = 2\hat{\sigma}$. White dots in (a) and (b) are the training points.

3.9 Effect of Sample Size

The learning curves are shown in Fig. 11. The polynomial stays at round-off ($\sim 10^{-12}$) at every N , because 20 points already over-determine its 15 coefficients. The GP improves fastest, from a median grid RMSE of 2.91 at $N = 20$ to 0.029 at $N = 45$ and 0.0098 at $N = 90$: going from 20 to 45 points buys a factor of 100, going from 45 to 90 only a factor of three. The RBF improves slowly, from 162 to 27.5 over the same range; in the central box it falls faster, from 61 to 3.8. Sampling variability is large at small N : at $N = 30$ the five GP runs span a factor of 19 (0.093 to 1.79), which is a strong reason not to report a single-seed number as “the” accuracy of a model.

3.10 Surrogate-Based Optimization and Verification

Table 7 and Fig. 12 give the optimization results. All 20 starts on the polynomial converge to $(1.000000, 1.000000)$, and the true function there is 7.5×10^{-16} . All 20 starts on the GP converge to $(0.9940, 0.9883)$, only 0.013 from the true minimum, with a true value of 4.5×10^{-5} against a predicted 6.1×10^{-4} . The GP’s positional error, -0.0060 in x_1 and -0.0117 in x_2 , has a slope of 1.95. That is almost exactly the slope of the valley tangent, $2x_1 = 2$ at $x_1 = 1$. The surrogate found the valley, but not the precise point along its nearly flat floor, which is exactly the difficulty that makes Rosenbrock a benchmark.

The RBF shows why verification is not optional. Its best candidate, reached by 11 of the 20 starts, is a *spurious* minimum at $(-0.44, 0.04)$ with a predicted $\hat{f} = -16.8$: negative, although $f \geq 0$ everywhere, and 1.73 away from the true minimum, where the true value is 4.50. It is an

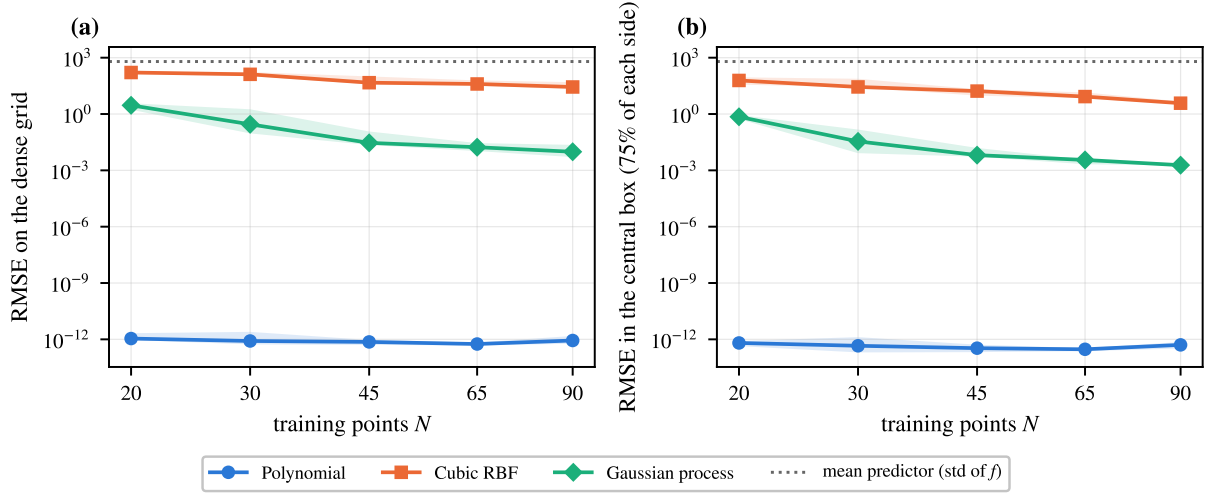


Figure 11: Rosenbrock: learning curves on logarithmic axes. (a) RMSE over the full 120×120 grid. (b) RMSE over the central box whose sides are 75% of the domain sides. Lines are medians over five independent LHS designs per training size and shaded bands span their minimum and maximum; the dotted line is the error of predicting the mean (standard deviation of f on the grid, 624).

undershoot of the interpolant between samples inside the valley. The candidate closest to the truth, $(0.94, 0.87)$ with $f = 0.031$, ranks only second on the surrogate. Trusting the surrogate's own ranking would have picked the wrong design; one true evaluation per candidate, six in total, exposes the error. A second, small undershoot patch near $(1.2, 1.45)$ ($\hat{f} < -3.6$, Fig. 12b) was not reached by any start, so the multi-start search found four of at least five RBF minima, a reminder that 20 starts do not guarantee a complete search.

Table 7: Surrogate-based optimization (20 bounded Nelder–Mead starts per surrogate) and verification with the true function. “Hits” is the number of starts that ended in each cluster; d is the distance to the known minimum. The last row of each block is the same multi-start search on the true function.

Function	Surrogate	Hits	$\hat{\mathbf{x}}^*$	Predicted $\hat{f}(\hat{\mathbf{x}}^*)$	True $f(\hat{\mathbf{x}}^*)$	$ \hat{f} - f $	d
Rosenbrock	Polynomial	20	(1.0000, 1.0000)	-5.0×10^{-13}	7.5×10^{-16}	5.1×10^{-13}	2.1×10^{-9}
	RBF, cluster 1	11	(-0.4388, 0.0366)	-16.77	4.503	21.27	1.732
	RBF, cluster 2	3	(0.9425, 0.8718)	-1.761	0.0307	1.792	0.140
	RBF, cluster 3	3	(0.7209, 0.4026)	-0.797	1.449	2.247	0.659
	RBF, cluster 4	3	(-1.1492, 1.3781)	8.312	4.948	3.364	2.182
	GP	20	(0.9940, 0.9883)	6.1×10^{-4}	4.5×10^{-5}	5.7×10^{-4}	1.3×10^{-2}
	True f (check)	20	(1.0000, 1.0000)	–	1.9×10^{-20}	–	2.6×10^{-10}
Trid	Polynomial	20	(2.0000, 2.0000)	-2.000000	-2.000000	1.3×10^{-14}	1.0×10^{-8}
	RBF	20	(1.9950, 1.9933)	-2.012319	-1.999964	1.2×10^{-2}	8.4×10^{-3}
	GP	20	(1.9995, 1.9997)	-2.000000	-2.000000	2.7×10^{-7}	5.7×10^{-4}
	True f (check)	20	(2.0000, 2.0000)	–	-2.000000	–	1.1×10^{-8}

3.11 Extrapolation

Table 8 and Fig. 13 give the reduced-box experiment (training on $[-1, 1]^2$). Inside the box the GP error is small (RMSE 0.0064); outside it rises to 14.9, a factor of 2320, with a worst error of 121 in the corners, where the true function reaches 3609 but the GP has no data to follow the quartic growth. The RBF, whose linear tail extrapolates linearly, is worse in absolute terms (574 outside). The polynomial stays exact outside the box, but only because its assumed form happens to be the true one. In a real problem that is unknowable, and a surrogate should never

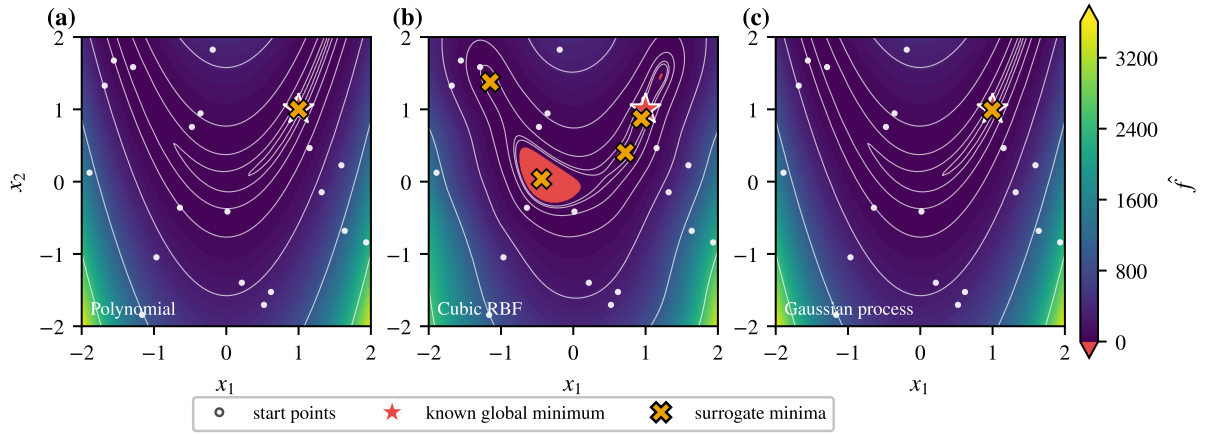


Figure 12: Rosenbrock: multi-start optimization on (a) the degree-4 polynomial, (b) the cubic RBF and (c) the GP. The background is each surrogate $\hat{f}$ with the color limits of the true function and white iso-lines of $\hat{f}$; white dots are the 20 start points, crosses the distinct surrogate minima, and the star the known global minimum. Four distinct RBF minima were found from the 20 starts, including the spurious deepest one at $(-0.44, 0.04)$, which lies inside the large red region where $\hat{f} < -3.6$. The small red patch near $(1.2, 1.45)$ is a further undershoot that no start reached.

be trusted outside its training box. Note also that the true minimum $(1, 1)$ lies on the corner of this training box, so any optimizer that stops at the edge of the DOE box is telling the engineer to re-sample, not to trust the answer.

Table 8: Extrapolation test: each model is trained on 45 LHS points in the central sub-box $([-1, 1]^2$ for Rosenbrock, $[-2, 2]^2$ for Trid) and evaluated over the full grid. The last column is the ratio of the RMSE outside to the RMSE inside the training box.

Function	Model	RMSE inside	RMSE outside	MaxAE inside	MaxAE outside	Ratio out/in
Rosenbrock	Polynomial	5.9×10^{-14}	8.3×10^{-13}	2.0×10^{-13}	3.6×10^{-12}	14
	Cubic RBF	6.07	574.2	66.4	2622	95
	GP	6.4×10^{-3}	14.93	0.101	120.6	2321
Trid	Polynomial	2.8×10^{-15}	1.2×10^{-14}	8.9×10^{-15}	4.3×10^{-14}	4.4
	Cubic RBF	5.3×10^{-2}	3.712	0.831	19.12	70
	GP	9.9×10^{-6}	2.8×10^{-3}	1.7×10^{-4}	2.6×10^{-2}	287

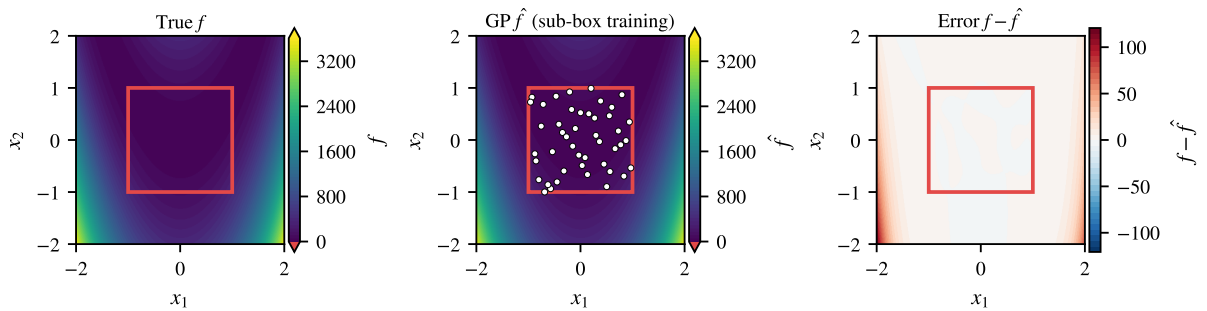


Figure 13: Rosenbrock, extrapolation of a GP trained only inside the red box $[-1, 1]^2$ (white dots). Left: true f . Center: GP prediction over the full domain with the color limits of the true function (values above them saturate; no value falls below the red undershoot limit). Right: error $f - \hat{f}$, which grows by three orders of magnitude outside the training box.

4 Results, Part B: Trid Function

4.1 Function, Domain and Analytical Minimum

The two-variable Trid function, Eq. (3), is shown in Fig. 14. Its minimum can be found by hand. Setting the gradient to zero,

$$\nabla f = \begin{bmatrix} 2(x_1 - 1) - x_2 \\ 2(x_2 - 1) - x_1 \end{bmatrix} = \mathbf{0} \implies \underbrace{\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}}_{\mathbf{H}} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}. \quad (15)$$

Adding the two equations gives $x_1 + x_2 = 4$; subtracting them gives $3(x_1 - x_2) = 0$. Hence

$$\mathbf{x}^* = (2, 2), \quad f(\mathbf{x}^*) = (2 - 1)^2 + (2 - 1)^2 - 2 \cdot 2 = -2. \quad (16)$$

The Hessian $\mathbf{H}$ is constant, with eigenvalues 1 and 3 and eigenvectors along $(1, 1)$ and $(1, -1)$. It is positive definite, so f is strictly convex and $(2, 2)$ is its unique global minimum on any domain containing it. The level sets are ellipses tilted at 45° with axis ratio $\sqrt{3}$, elongated along $(1, 1)$, the direction of the smaller eigenvalue (Fig. 14a). On $[-4, 4]^2$ the function ranges from -2 to 50, the maximum occurring at the corners $(-4, 4)$ and $(4, -4)$, where the cross term $-x_1x_2$ is largest and positive.

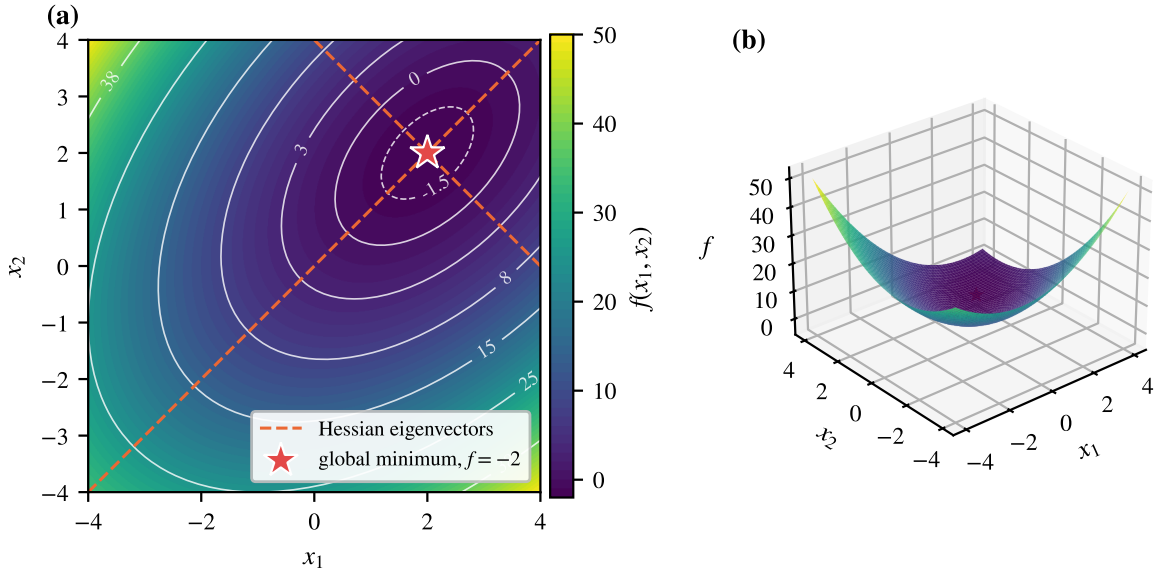


Figure 14: Trid function on $[-4, 4]^2$. (a) Filled contour map with iso-lines at $f = -1.5, 0, 3, 8, 15, 25, 38$; dashed lines are the Hessian eigenvector directions $(1, 1)$ and $(1, -1)$ through the minimum, and the star is the global minimum $f = -2$ at $(2, 2)$. (b) Three-dimensional surface with the minimum marked.

4.2 Design of Experiments and Split

The Trid study uses the same maximin LHS as Part A (Fig. 2c and Table 2), scaled to $[-4, 4]^2$. Its $\phi_{\min} = 0.0452$ in unit coordinates corresponds to 0.36 in physical units, and the characteristic spacing $\sqrt{|\mathcal{X}|/n}$ of the 45 training points is 1.19. That spacing is adequate here because the Trid function has no short-wavelength features: its curvature is constant, and a quadratic is fully determined by six well-placed values. The same argument justifies $N = 60$: ten times the six coefficients of the exact model, with 36 points per CV fold. The split is shown in Fig. 15; the test points cover all four quadrants, and two of them, $(1.83, 2.72)$ and $(2.91, 1.77)$, lie 0.74 and 0.94 from the minimum.

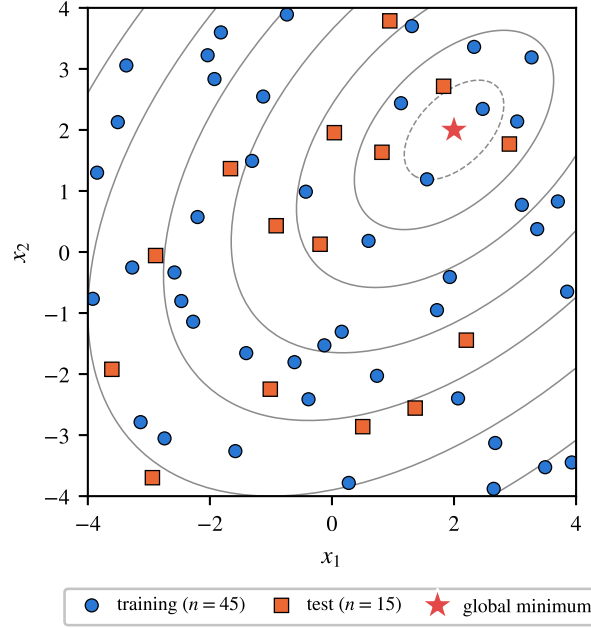


Figure 15: Trid: locations of the 45 training points and the 15 held-out test points over iso-lines of f ; the star is the global minimum.

4.3 Polynomial Degree Selection

The Trid degree sweep (Fig. 16, Table 3) repeats the Rosenbrock pattern at a lower degree. Degree 1 under-fits: a plane cannot represent a bowl, and its train, CV and test RMSE (9.68, 10.1 and 6.22) are almost equal to the spread of the data itself, a pure bias error. At $p = 2$ the CV RMSE drops to 1.04×10^{-14} , 15 orders of magnitude lower, and the parsimony rule selects the true degree. Degrees 3 to 6 stay at round-off, degree 7 rises to 5.6×10^{-12} with the condition number, and degree 8 jumps to 0.92 because the folds are under-determined, as explained for Part A. The recovered physical coefficients match Eq. (3) to within 1.2×10^{-14} . Solving $\nabla \hat{f} = \mathbf{0}$ for the *fitted* quadratic gives a stationary point that differs from (2, 2) by 1.0×10^{-14} in x_1 and 5×10^{-15} in x_2 , and a Hessian equal to $\mathbf{H}$ of Eq. (15) to within 4×10^{-15} , reproducing the hand derivation of Eq. (16) from the surrogate itself.

4.4 Held-Out Validation, Cross-Validation and GP Hyperparameters

The Trid metrics are in Tables 4 and 5, and the parity plots in Fig. 17. All models beat the baseline (test RMSE 9.73, $R^2 = -0.26$). The polynomial is exact to 7.9×10^{-15} ; the GP reaches 2.9×10^{-5} (NRMSE 1.0×10^{-6}); and the RBF reaches 0.042 (NRMSE 1.5×10^{-3}) with a test MaxAE of 0.155. As for Rosenbrock, the RBF test RMSE underestimates the grid RMSE (0.156) by a factor of almost four, while CV (0.178) is close to it.

The GP length scales are 37.5 and 37.2 in physical units (Table 6), almost identical and about 4.7 times the box width. Equal length scales are expected: the Trid function is symmetric under $x_1 \leftrightarrow x_2$ and so is the design box. Their size says that the response is low-order and smooth relative to the domain, so the GP is effectively modeling a polynomial-like trend. As for Rosenbrock, the signal variance sits at its upper bound and the noise variance near its lower bound (1.9×10^{-12}); here 8 of the 9 optimizer runs ended abnormally. Widening the bound gave lower likelihoods (260.1 and 254.6 against 302.6), which again reveals other local optima rather than better models, and worse CV errors (Table 6), so the 10^4 bound was kept for the reason given in Sec. 3.6.

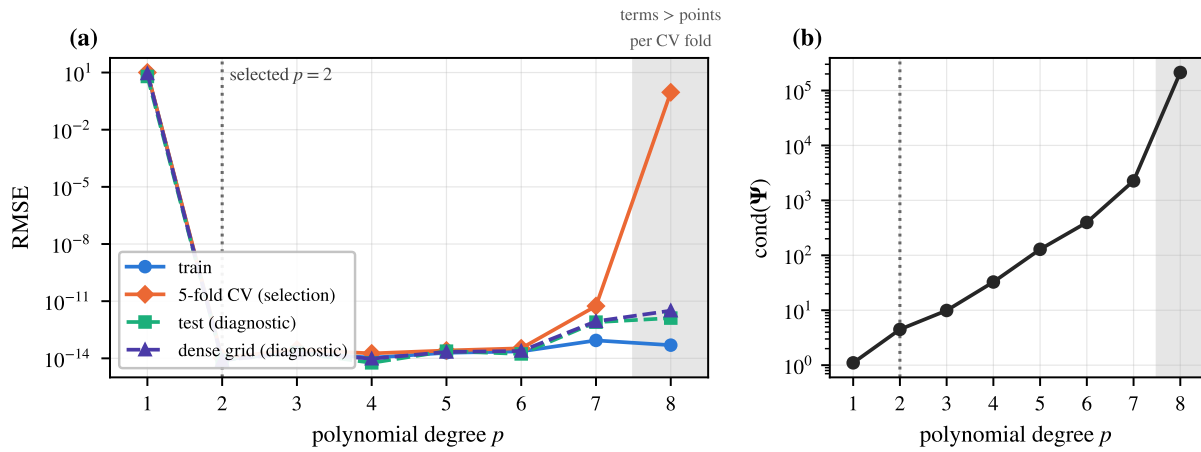


Figure 16: Trid: polynomial complexity study. (a) Training, five-fold CV (selection) and test and dense-grid (diagnostic) RMSE against degree; the dotted line marks the selected $p = 2$ and the shaded band marks degrees with more coefficients than points per CV fold. (b) Condition number of the standardized design matrix.

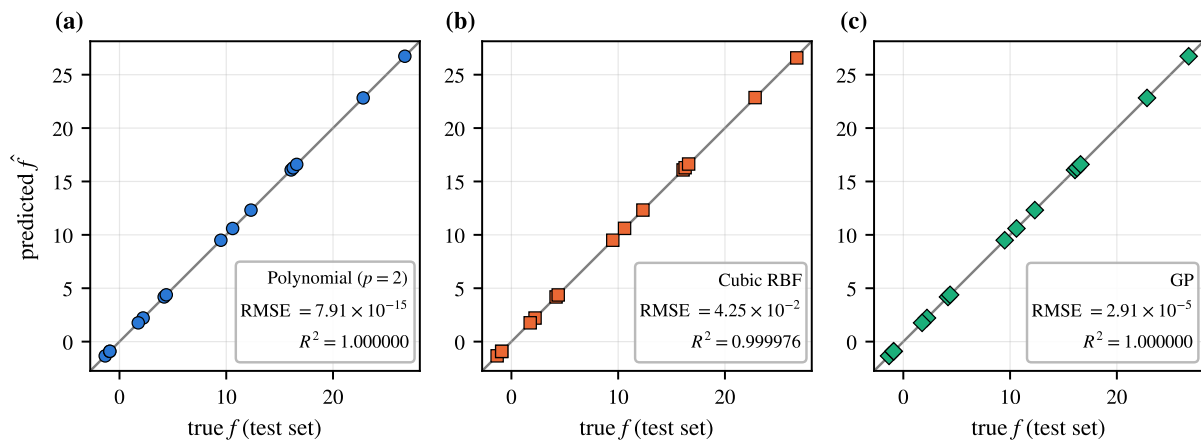


Figure 17: Trid: parity plots on the 15 held-out test points for (a) the degree-2 polynomial, (b) the cubic RBF and (c) the GP, with test RMSE and R^2 in the insets.

4.5 Side-by-Side Visual Comparison

The side-by-side comparisons are in Figs. 18 to 20, and the 3-D pair of the best surrogate in Fig. 21. The polynomial error is round-off (2.8×10^{-14}). The GP error (Fig. 19) is below 10^{-4} in the interior and peaks at 9.6×10^{-4} at the edges and corners. The RBF error (Fig. 20) is the clearest failure pattern of the whole study: it is small inside the sampled region (the learning curve of Fig. 23 gives a central-box RMSE of about 0.03 at $N = 45$) and grows sharply toward the boundary, reaching 2.72 in the corner $(-4, 4)$. The Trid function has constant curvature everywhere; the cubic RBF with a linear tail cannot represent a quadratic exactly, and near the boundary, where no samples lie beyond, it relaxes toward its linear tail and loses the curvature [19]. The error is therefore largest precisely where the true surface is steepest and the data give no support from outside.

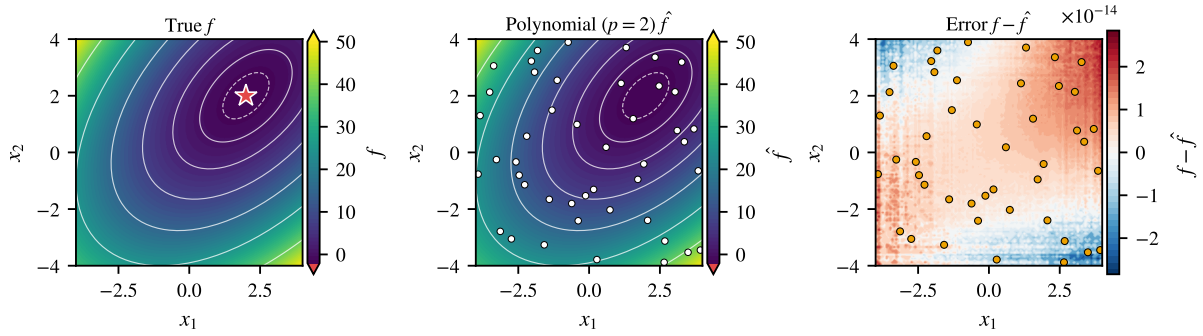


Figure 18: Trid, best surrogate by CV (degree-2 polynomial): true f , surrogate $\hat{f}$ with identical color limits and training points, and error $f - \hat{f}$ on a diverging scale centered at zero with training points (maximum $|e| = 2.8 \times 10^{-14}$, round-off).

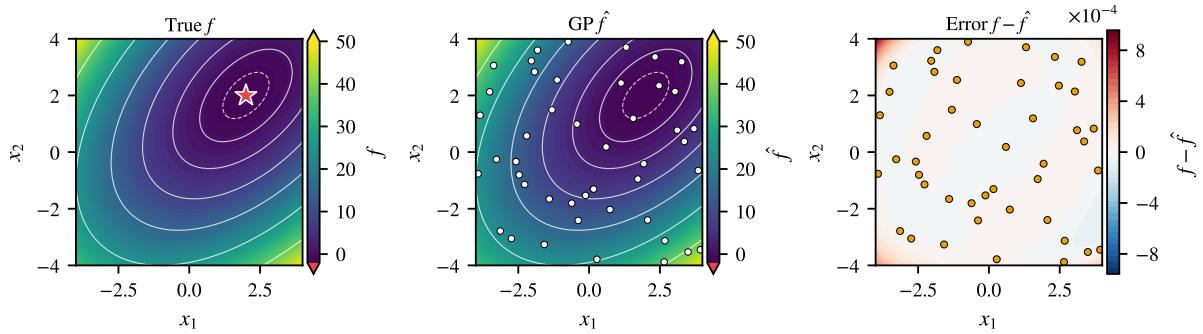


Figure 19: Trid, GP surrogate: true f , surrogate $\hat{f}$ with identical color limits, and error $f - \hat{f}$ (diverging scale centered at zero, maximum $|e| = 9.6 \times 10^{-4}$), with training points overlaid.

4.6 Uncertainty Quantification

For the Trid GP (Fig. 22), $\hat{\sigma}$ is again smallest near the samples and largest at the edges and corners. Every grid point satisfies $|f - \hat{f}| \leq 2\hat{\sigma}$, so the uncertainty band is conservative. The Pearson correlation is 0.68, but the Spearman rank correlation is only 0.27. The reason is visible in panel (b): in the interior the true error is at the level of 10^{-5} to 10^{-8} and forms thin curved bands where $f - \hat{f}$ changes sign, a structure that has nothing to do with distance from the data. $\hat{\sigma}$ tracks the large errors at the edges (hence the reasonable Pearson value), but in the interior it is itself at its numerical floor: its median over the central box, 7.5×10^{-5} , is of the

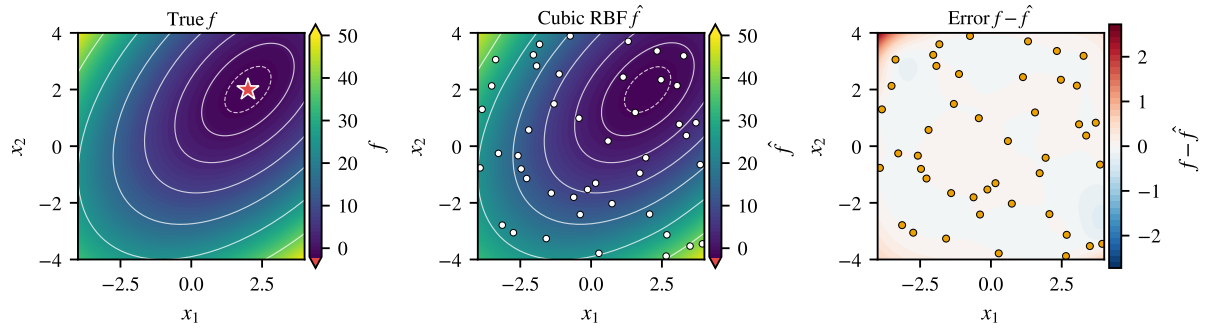


Figure 20: Trid, cubic RBF surrogate: true f , surrogate $\hat{f}$ with identical color limits, and error $f - \hat{f}$ (diverging scale centered at zero, maximum $|e| = 2.72$ in the corner $(-4, 4)$), with training points overlaid. The error is confined to a band along the boundary.

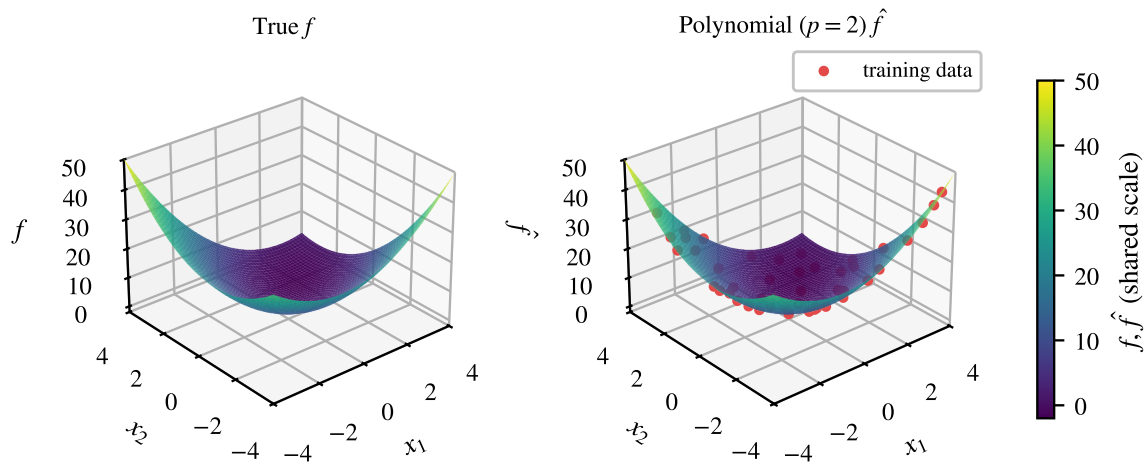


Figure 21: Trid: 3-D surfaces of the true function and of the degree-2 polynomial surrogate with identical axis limits, color limits and viewing angle; red dots are the training data.

same order as the floor $\sqrt{\alpha + \sigma_n^2} \text{std}(y_{\text{train}}) = 1.2 \times 10^{-4}$ and as its value at the training points, 1.1×10^{-4} . Both $\hat{\sigma}$ and the true error are therefore dominated by numerical effects there, and one cannot rank the other. A rerun of the same script with the same package versions on a Linux machine confirms this: every other result in this report agrees within 1%, but the Trid correlations change to 0.20 (Spearman) and 0.65 (Pearson), because they are computed from quantities at round-off level. This is a clear illustration that $\hat{\sigma}$ is a statement about the model's assumptions and the sample geometry, not a measurement of the actual error.

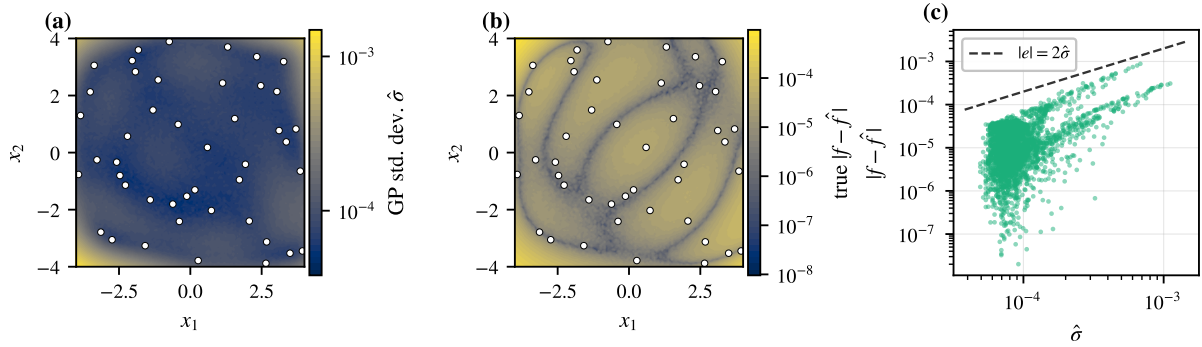


Figure 22: Trid, GP uncertainty. (a) Predicted standard deviation $\hat{\sigma}$ and (b) true absolute error $|f - \hat{f}|$, both on log color scales with the training points. (c) $|f - \hat{f}|$ against $\hat{\sigma}$ for 3000 random grid points; every point lies below the line $|e| = 2\hat{\sigma}$.

4.7 Effect of Sample Size

The Trid learning curves (Fig. 23) separate interior from edge behavior. In the central box the RBF error falls steadily with N , from 0.235 to 0.0094, as a converging interpolant should. Over the full grid, however, its median RMSE stops falling after $N = 45$ (0.091, then 0.110 at $N = 65$ and 0.177 at $N = 90$). With five draws per size the ranges overlap (0.069 to 0.218 at $N = 45$, 0.117 to 0.265 at $N = 90$), so the rise itself is not significant, but the stagnation against a falling core error is clear. The median condition number of the saddle-point system grows from 1.0×10^4 at $N = 20$ to 2.2×10^6 at $N = 90$, far from causing round-off trouble in double precision, so conditioning is not the explanation. A plausible one is the boundary behavior of polyharmonic splines [19]: added interior points improve the interior but do little for the loss of curvature in the edge band, which dominates the full-grid error. The GP shows the same pattern at a much lower level: its full-grid median rises slightly (from 4.8×10^{-5} at $N = 45$ to 7.0×10^{-5} at $N = 90$) while its core error settles at about 1.7×10^{-5} . That floor is most likely set by the near-degenerate kernel (very long length scales and a noise variance near 10^{-12}), not by the amount of data. The polynomial is at round-off for every N .

4.8 Surrogate-Based Optimization and Verification

All three surrogates lead every one of the 20 starts to a single minimum (Table 7 and Fig. 24). The polynomial minimum lies 1.0×10^{-8} from $(2, 2)$, a distance set by the Nelder–Mead tolerance, with a true value of -2.000000 . The GP minimum is 0.00057 from $(2, 2)$ and its true value, -1.9999998 , agrees with the prediction to 2.7×10^{-7} . The RBF minimum is 0.0084 from $(2, 2)$, and it predicts $\hat{f} = -2.0123$, a value *lower* than the true global minimum -2 : the surrogate promises a design better than any design that exists. The true value at its point, -1.99996 , is still within 3.6×10^{-5} of the optimum, so for this convex function the positional error is harmless, but the predicted performance is optimistic. The multi-start search on the true

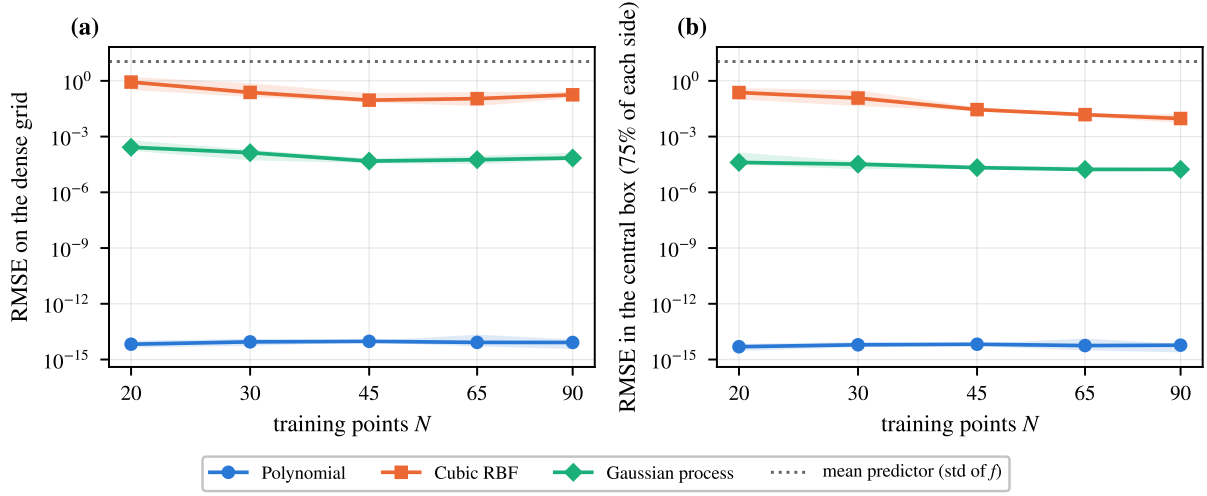


Figure 23: Trid: learning curves on logarithmic axes. (a) RMSE over the full grid. (b) RMSE over the central box whose sides are 75% of the domain sides. Lines are medians over five LHS designs per size and bands their range; the dotted line is the error of predicting the mean (standard deviation of f , 10.9). Beyond $N = 45$ the full-grid RBF and GP errors stop falling, while their core errors do not rise.

function returns $(2, 2)$ and $f = -2$, confirming the hand derivation of Eq. (16) numerically. The analytical solution, the stationary point of the fitted quadratic and the numerical searches all agree.

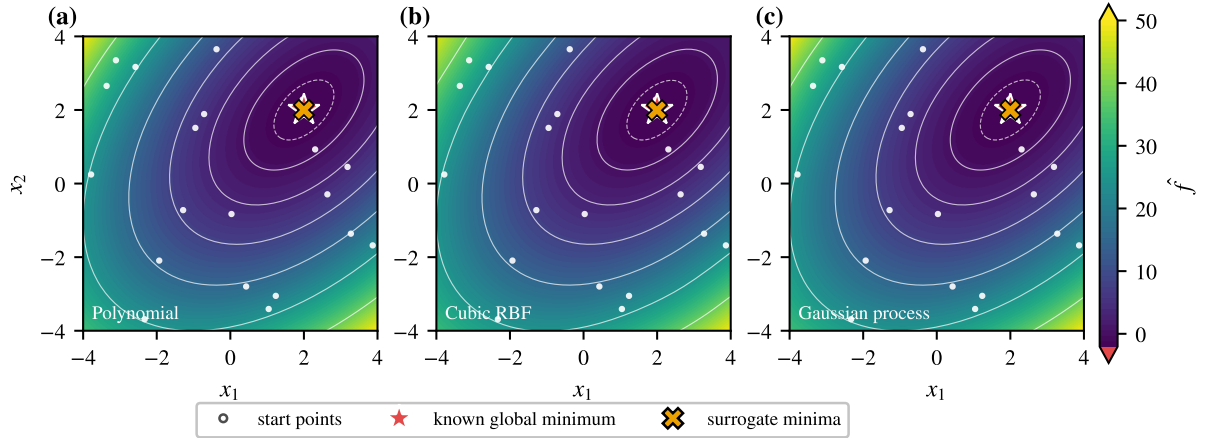


Figure 24: Trid: multi-start optimization on (a) the degree-2 polynomial, (b) the cubic RBF and (c) the GP, drawn over each surrogate with the color limits of the true function; white dots are the 20 start points, crosses the surrogate minima and the star the known global minimum at $(2, 2)$.

4.9 Extrapolation

Trained on $[-2, 2]^2$, which has the minimum $(2, 2)$ at its corner, the GP error rises from 9.9×10^{-6} inside to 2.8×10^{-3} outside, a factor of 287 (Table 8 and Fig. 25). The ratio is smaller than for Rosenbrock because the Trid function keeps the same quadratic shape everywhere, so the long-length-scale GP continues the trend reasonably well. The RBF error grows by a factor of 70, to 19 at the corners, because its linear tail cannot follow the quadratic growth. The polynomial again extrapolates exactly, for the same reason as in Part A.

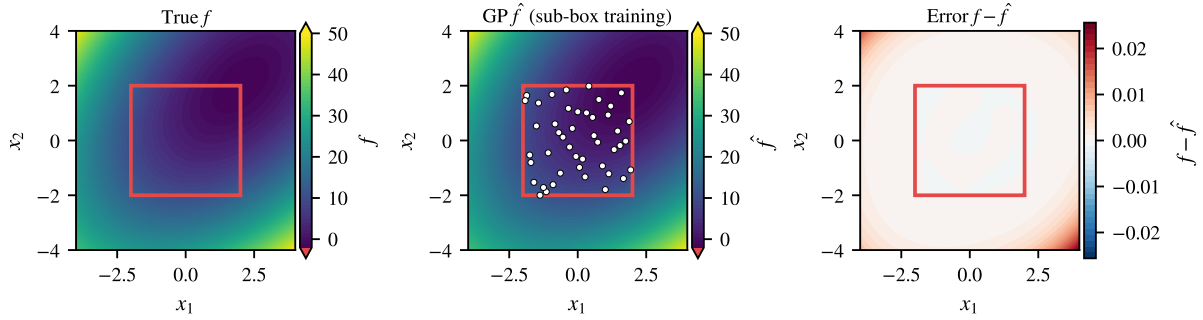


Figure 25: Trid, extrapolation of a GP trained only inside the red box $[-2, 2]^2$ (white dots): true f , GP prediction with the color limits of the true function, and error $f - \hat{f}$, which grows toward the corners of the full domain.

4.10 Evaluation Budget

Table 9 separates the main DOE budget from the extra evaluations made only because the truth is cheap. Only the 60 DOE runs and the verification runs would exist in a real project; the grid, learning-curve, reduced-box and true-function multi-start evaluations are teaching diagnostics and were never used for fitting or selection.

Table 9: Accounting of true-function evaluations. Only the first two rows would be spent in a real study.

Purpose	Rosenbrock	Trid
Main DOE (45 training + 15 test)	60	60
Verification of surrogate optima	6	3
Dense truth grid (diagnostic)	14 400	14 400
Learning curves (5 sizes $\times$ 5 draws)	1250	1250
Reduced-box extrapolation training	45	45
Multi-start check on the true function	4424	3315

5 Discussion

5.1 Comparing the Two Functions

Both functions are exact polynomials, so the polynomial surrogate has zero bias and wins every comparison by nine or more orders of magnitude. That is not a fair fight, and it is not the main lesson. In a real project the true form is unknown. What the data *do* show is a collapse of the CV error by 14 to 15 orders of magnitude at one particular degree, followed by a flat plateau at round-off. With noiseless simulation data such a collapse is strong evidence that the response is an exact low-degree polynomial; with noisy or non-polynomial data it would not occur, and the CV curve would instead have a shallow minimum.

The two functions differ sharply in difficulty for the non-polynomial surrogates. On a common scale (NRMSE, Table 4), the RBF is 12 times worse on Rosenbrock (1.9×10^{-2}) than on Trid (1.5×10^{-3}), and the GP about 7 times worse. The Rosenbrock function has a dynamic range of 3609 concentrated in the corners, quartic growth that a linear-tail RBF cannot follow, and a curved valley that axis-aligned length scales cannot describe. The Trid function has constant curvature, a range of 52 and a symmetry that the GP recovers as two equal length scales.

The practical consequence is visible in the optimization: the same RBF that places the Trid minimum within 0.008 creates a false, negative minimum in the Rosenbrock valley.

5.2 Catalog Prompts

Rosenbrock. The GP length scales, 3.11 for x_1 and 27.4 for x_2 , correctly rank x_1 as the more influential variable. They cannot describe the valley itself, whose direction rotates across the domain. The “easy to find the valley, hard to find the minimum” character appears directly in Table 7: the GP optimum is displaced by 0.013 almost exactly along the valley tangent, at a true cost of only 4.5×10^{-5} , because the function is nearly flat in that direction. A tiny error in f coexists with a visible error in $\mathbf{x}$.

Trid. Degree 2 is exact, and CV selects it from the data. The minimum derived by hand, $(2, 2)$ with $f = -2$ from the 2×2 system of Eq. (15), is reproduced by the stationary point of the fitted quadratic to 10^{-14} , by the multi-start searches on the polynomial and the GP (within 10^{-8} and 6×10^{-4}), approximately by the RBF (within 0.008), and by the same search on the true function. The positive-definite Hessian, with eigenvalues 1 and 3, proves that this stationary point is the unique global minimum, which no numerical search alone can prove.

5.3 Connection to Theory

The results follow the bias–variance picture of Sec. 2.5. Degrees below the true one fail through bias, with train, CV and test errors high and close together. Degrees above it show no variance-driven over-fitting, because the data are noiseless; they show numerical growth through the condition number, and under-determination when the CV folds have fewer points than coefficients. The RBF results match the known boundary behavior of polyharmonic splines [19]: interior accuracy converges with N , while the edge error stagnates. The GP results match the stationary-kernel picture [11]: accuracy is excellent where the data constrain the model, uncertainty grows away from the data, and the kernel drifts to its bounds when asked to represent a global polynomial trend.

5.4 Where Each Model Can Be Trusted

Table 10 summarizes the engineering judgment. The polynomial is trusted everywhere in both boxes, but that trust rests on the CV evidence of exact form. The GP is the model to use when the form is unknown: its interior accuracy is excellent, its optimum was verified within 0.013 (Rosenbrock) and 0.0006 (Trid), and $\hat{\sigma}$ points to the corners as its weakest region, although for Rosenbrock the $2\hat{\sigma}$ band underestimates the largest errors there. The RBF is acceptable for Trid trends and optimization in the interior, but not for Rosenbrock optimization, where it produces a spurious minimum with a physically impossible negative value. No model should be used outside its training box.

5.5 Limitations

Every result comes from one DOE and one split (seed 7); the learning curves show that single-seed errors can vary by an order of magnitude at small N , so the reported numbers should be read as one realization rather than as exact accuracies. Both functions are smooth polynomials; the conclusions about the polynomial surrogate would not transfer to non-polynomial responses. The GP uses a stationary squared-exponential kernel whose signal variance sits at its bound; universal Kriging or a different kernel was not explored. Finally, the design is one-shot: in a real study the verified optimum candidates would be added to the DOE and the surrogate refitted, for example with expected-improvement infill [12].

Table 10: Engineering judgment: which model to trust, for what purpose, and inside which bounds.

Function	Model	Recommended use and bounds
Rosenbrock	Polynomial, $p = 4$	Any purpose on $[-2, 2]^2$; exact form confirmed by the CV collapse.
	GP	Global trends and locating the optimum inside $[-2, 2]^2$ (verified $d = 0.013$); treat the corner regions, especially near $(-2, -2)$, with caution (error up to 0.70).
	Cubic RBF	Coarse trends only (test NRMSE 1.9%); not for optimization, because it creates spurious minima with $\hat{f} < 0$ in the valley.
Trid	Polynomial, $p = 2$	Any purpose on $[-4, 4]^2$; reproduces the analytical minimum and Hessian to round-off.
	GP	Any purpose on $[-4, 4]^2$ (NRMSE 10^{-6} , optimum within 6×10^{-4}).
	Cubic RBF	Interior of the box ($ x_j \lesssim 3$) and locating the optimum ($d = 0.008$); its predicted optimal value is optimistic by 0.012, and errors reach 2.7 at the corners.

6 Conclusions

A complete surrogate study was carried out on the Rosenbrock and Trid functions with a budget of 60 evaluations from a maximin Latin hypercube design, whose minimum pairwise distance is almost twice that of random sampling at the same cost. The main findings are:

1. Training-only cross-validation recovered the true polynomial degrees (4 and 2) from the data, and the fitted coefficients matched the exact expansions to within 4×10^{-13} . Degrees above the true one showed no noise-driven over-fitting, but they did show conditioning growth and under-determination inside the CV folds.
2. The GP was the best model that does not assume the true form, with test NRMSE of 7.8×10^{-6} (Rosenbrock) and 1.0×10^{-6} (Trid). Its uncertainty $\hat{\sigma}$ correlated with the true error (Pearson 0.70 and 0.68) and bounded it within $2\hat{\sigma}$ at 98% and 100% of grid points. For Rosenbrock, however, the points outside the band include the worst error, and for Trid $\hat{\sigma}$ could not rank errors near the numerical floor.
3. The cubic RBF failed systematically at the domain boundaries. For Trid its full-grid error stopped improving beyond $N = 45$ while its interior error kept falling. For Rosenbrock it produced a spurious negative minimum, and only verification with the true function exposed it.
4. Surrogate optima were verified with one true evaluation each. The GP located the Rosenbrock minimum within 0.013, displaced along the flat valley, and the Trid minimum within 6×10^{-4} . The hand-derived Trid minimum $f = -2$ at $(2, 2)$ was confirmed analytically, from the fitted quadratic, and numerically.
5. Outside a reduced training box, the RBF and GP errors grew by factors of 70 to 2300. A surrogate is a statement about the region it has seen, and must not be used outside it.

A cheap model can guide a design decision reliably, but only inside its sampled box, only after its prediction has been checked against the expensive model, and only when the engineer can show where it fails.

Acknowledgments

The workflow, the Latin hypercube and cubic RBF implementations, and the GP set-up are adapted from the AE 541 reference notebook on the Himmelblau function [6]; all functions, domains, settings, results and interpretations are the author’s own for the Rosenbrock and Trid functions. In line with Sec. 7 of the assignment brief, an AI assistant (Claude, Anthropic) was used to help structure the Python script and the L^AT_EX report, to review figure layout, and to improve the clarity of the English. Every equation, number and statement was checked by the author against the output of the submitted script, and the author takes full responsibility for the content.

References

- [1] Nestor V. Queipo, Raphael T. Haftka, Wei Shyy, Tushar Goel, Rajkumar Vaidyanathan, and P. Kevin Tucker. Surrogate-based analysis and optimization. *Progress in Aerospace Sciences*, 41(1):1–28, 2005. doi: 10.1016/j.paerosci.2005.02.001.
- [2] Alexander I. J. Forrester, András Sóbester, and Andy J. Keane. *Engineering Design via Surrogate Modelling: A Practical Guide*. Wiley, Chichester, UK, 2008. doi: 10.1002/9780470770801.
- [3] AE 541 Course Team. Assignment 1: Surrogate model development for two-variable test functions. AE 541 Digital Twin in Modern Engineering, King Fahd University of Petroleum and Minerals, 2026.
- [4] Howard H. Rosenbrock. An automatic method for finding the greatest or least value of a function. *The Computer Journal*, 3(3):175–184, 1960. doi: 10.1093/comjnl/3.3.175.
- [5] Sonja Surjanovic and Derek Bingham. Virtual library of simulation experiments: Test functions and datasets. Simon Fraser University, <https://www.sfu.ca/~ssurjano/>, 2013. Accessed September 2026.
- [6] AE 541 Course Team. From an expensive function to a trustworthy surrogate: A complete, step-by-step study using the Himmelblau function. Jupyter notebook `himmelblau_surrogate_tutorial.ipynb`, AE 541 Digital Twin in Modern Engineering, King Fahd University of Petroleum and Minerals, 2026.
- [7] Michael D. McKay, Richard J. Beckman, and William J. Conover. A comparison of three methods for selecting values of input variables in the analysis of output from a computer code. *Technometrics*, 21(2):239–245, 1979. doi: 10.2307/1268522.
- [8] Mark E. Johnson, Leslie M. Moore, and Donald Ylvisaker. Minimax and maximin distance designs. *Journal of Statistical Planning and Inference*, 26(2):131–148, 1990. doi: 10.1016/0378-3758(90)90122-B.
- [9] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [10] Martin D. Buhmann. *Radial Basis Functions: Theory and Implementations*. Cambridge University Press, Cambridge, UK, 2003. doi: 10.1017/CBO9780511543241.

-
- [11] Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, Cambridge, MA, 2006.
 - [12] Donald R. Jones, Matthias Schonlau, and William J. Welch. Efficient global optimization of expensive black-box functions. *Journal of Global Optimization*, 13(4):455–492, 1998. doi: 10.1023/A:1008306431147.
 - [13] Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, New York, 2nd edition, 2009. doi: 10.1007/978-0-387-84858-7.
 - [14] John A. Nelder and Roger Mead. A simplex method for function minimization. *The Computer Journal*, 7(4):308–313, 1965. doi: 10.1093/comjnl/7.4.308.
 - [15] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, et al. Array programming with NumPy. *Nature*, 585:357–362, 2020. doi: 10.1038/s41586-020-2649-2.
 - [16] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, et al. SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020. doi: 10.1038/s41592-019-0686-2.
 - [17] John D. Hunter. Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3):90–95, 2007. doi: 10.1109/MCSE.2007.55.
 - [18] Wes McKinney. Data structures for statistical computing in Python. In *Proceedings of the 9th Python in Science Conference*, pages 56–61, 2010. doi: 10.25080/Majora-92bf1922-00a.
 - [19] Bengt Fornberg, Tobin A. Driscoll, Grady Wright, and Richard Charles. Observations on the behavior of radial basis function approximations near boundaries. *Computers & Mathematics with Applications*, 43(3–5):473–490, 2002. doi: 10.1016/S0898-1221(01)00299-1.

A Complete Python Code

The listing below is the complete file 202510630_Bahaa_Karawia.py; running it reproduces every figure, table and number in this report.

```

1  """
2  AE 541 - Digital Twin in Modern Engineering
3  Assignment 1: Surrogate model development for two-variable test functions
4
5  Author : Bahaa Karawia (KFUPM ID 202510630)
6  Part A : Rosenbrock function (common)
7  Part B : Trid function, 2 variables (assigned)
8
9  The workflow follows the course reference notebook
10 (himmelblau_surrogate_tutorial.ipynb); the LHS, cubic RBF and GP set-up are
11 adapted from it with acknowledgment.
12
13 Run:  python 202510630_Bahaa_Karawia.py
14 Needs: numpy, scipy, matplotlib, scikit-learn, pandas.
15 Every figure and table in the report is written to ./outputs next to this file.
16 """
17
18 import json
19 import platform
20 import warnings
21 from pathlib import Path
22
23 import matplotlib
24 matplotlib.use("Agg") # write files only, no GUI window
25 import matplotlib.pyplot as plt
26 import numpy as np
27 import pandas as pd
28 import scipy
29 import sklearn
30 from matplotlib.patches import Rectangle
31 from matplotlib.colors import LogNorm
32 from matplotlib.ticker import LogLocator, MaxNLocator, ScalarFormatter
33 from scipy.optimize import minimize
34 from scipy.spatial.distance import cdist, pdist
35 from scipy.stats import spearmanr
36 from sklearn.base import BaseEstimator, RegressorMixin
37 from sklearn.exceptions import ConvergenceWarning
38 from sklearn.gaussian_process import GaussianProcessRegressor
39 from sklearn.gaussian_process.kernels import RBF, ConstantKernel, WhiteKernel
40 from sklearn.linear_model import LinearRegression
41 from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
42 from sklearn.model_selection import KFold, cross_val_predict, train_test_split
43 from sklearn.pipeline import make_pipeline
44 from sklearn.preprocessing import PolynomialFeatures, StandardScaler
45
46
47 # ===== CONFIGURATION =====
48 SEED = 7 # master seed; every random draw derives from it
49 SEED_DOE_STUDY = SEED + 1 # repeated designs for the random-vs-LHS comparison
50 SEED_STARTS = SEED + 2 # start points of the multi-start optimisation
51 SEED_SUB_BOX = SEED + 3 # design inside the reduced (extrapolation) box
52 SEED_STRIDE = 1000 # learning-curve draw r at size N uses SEED + 1000 r + N
53 N_SAMPLES = 60 # main DOE budget = training + held-out test
54 TEST_FRACTION = 0.25 # 45 training / 15 test points
55 CV_FOLDS = 5 # K-fold CV, applied to the training set only
56 POLY_DEGREES = range(1, 9) # degree sweep for the polynomial surrogate
57 PARSIMONY_TOL = 0.05 # degrees within 5 % of the best CV error count as tied
58 ROUNDOFF_REL = 1e-9 # CV RMSE below this x std(y_train) counts as exact
59 MAXIMIN_CANDIDATES = 100 # LHS candidates screened for the maximin design
60 DOE_REALISATIONS = 500 # repeated designs used to compare the DOE criteria
61 GRID_N = 120 # dense truth grid, GRID_N x GRID_N
62 GP_RESTARTS = 8 # restarts of the marginal-likelihood optimiser
63 GP_SIGNAL_MIN = 1e-3 # bounds on sigma_f^2 (outputs are normalised)
64 GP_SIGNAL_MAX = 1e4
65 GP_SIGNAL_TRIALS = [1e4, 1e6, 1e8] # upper bounds tried when sigma_f^2 hits its limit

```

```

66 GP_LENGTH_BOUNDS = (1e-2, 1e3)      # length scales, in standardised input units
67 GP_NOISE_BOUNDS = (1e-12, 1e-2)     # sigma_n^2; tiny because the data are noiseless
68 GP_INIT = {"signal": 1.0, "length": [1.0, 1.0], "noise": 1e-8} # optimiser start
69 GP_ALPHA = 1e-10                    # diagonal jitter for numerical stability (sklearn default);
70                                     # together with sigma_n^2 it sets a floor under sigma_hat
71 GP_BOUND_TOL = 1e-2                 # |log theta - log bound| below this counts as "at bound"
72 CORE_FRAC = 0.75                    # central fraction of the box used for "core" errors
73 LEARNING_N = [20, 30, 45, 65, 90]  # training sizes for the learning curve
74 LEARNING_REPEATS = 5                # independent LHS draws per training size
75 N_STARTS = 20                       # multi-start local searches on each surrogate
76 NM_XATOL, NM_FATOL = 1e-9, 1e-12    # Nelder-Mead stopping tolerances
77 NM_MAXITER = 5000
78 CLUSTER_FRAC = 0.02                 # minima closer than this x box width are merged
79 BOUNDARY_TOL = 1e-6                 # an optimum this close to a bound is flagged
80 SCATTER_POINTS = 3000               # grid points shown in the sigma-vs-error scatter
81 UNDERSHOOT_TOL = 1e-3               # surrogate values below min(f) by more than this x range
82                                     # of f are drawn in red on the field maps
83 OUTPUT_DIR = Path(__file__).resolve().parent / "outputs"
84
85
86 def rosenbrock(x1, x2):
87     return 100.0 * (x2 - x1**2) ** 2 + (1.0 - x1) ** 2
88
89
90 def trid(x1, x2):
91     return (x1 - 1.0) ** 2 + (x2 - 1.0) ** 2 - x1 * x2
92
93
94 FUNCTIONS = {
95     "rosenbrock": {
96         "name": "Rosenbrock",
97         "func": rosenbrock,
98         "bounds": np.array([[ -2.0, 2.0], [ -2.0, 2.0]]),
99         "minima": np.array([[1.0, 1.0]]),
100        "f_min": 0.0,
101        "sub_box": np.array([[ -1.0, 1.0], [ -1.0, 1.0]]),
102        # expanded form: 100x1^4 - 200x1^2x2 + 100x2^2 + x1^2 - 2x1 + 1
103        "exact_coefs": {"1": 1.0, "x1": -2.0, "x1^2": 1.0, "x2^2": 100.0,
104                        "x1^2 x2": -200.0, "x1^4": 100.0},
105        "line_levels": [0.5, 3, 15, 60, 250, 1000, 2500],
106        "view": (28, -118),
107    },
108    "trid": {
109        "name": "Trid",
110        "func": trid,
111        "bounds": np.array([[ -4.0, 4.0], [ -4.0, 4.0]]),
112        "minima": np.array([[2.0, 2.0]]),
113        "f_min": -2.0,
114        "sub_box": np.array([[ -2.0, 2.0], [ -2.0, 2.0]]),
115        # expanded form: x1^2 + x2^2 - x1x2 - 2x1 - 2x2 + 2
116        "exact_coefs": {"1": 2.0, "x1": -2.0, "x2": -2.0, "x1^2": 1.0,
117                        "x1 x2": -1.0, "x2^2": 1.0},
118        "line_levels": [-1.5, 0, 3, 8, 15, 25, 38],
119        "view": (28, -130),
120    },
121 }
122
123 MODEL_NAMES = ["Polynomial", "Cubic RBF", "Gaussian process"]
124 MODEL_COLORS = {"Polynomial": "#2a78d6", "Cubic RBF": "#eb6834",
125                 "Gaussian process": "#1baf7a"}
126 MODEL_MARKERS = {"Polynomial": "o", "Cubic RBF": "s", "Gaussian process": "D"}
127 # =====
128
129
130 def set_plot_style():
131     plt.rcParams.update({
132         "font.family": "serif",
133         "font.serif": ["STIXGeneral", "Times New Roman", "DejaVu Serif"],
134         "mathtext.fontset": "stix",
135         "font.size": 9,
136         "axes.labelsize": 9.5,

```

```

137     "xtick.labelsize": 8.5,
138     "ytick.labelsize": 8.5,
139     "legend.fontsize": 8,
140     "axes.linewidth": 0.8,
141     "xtick.direction": "out",
142     "ytick.direction": "out",
143     "lines.linewidth": 1.5,
144     "lines.markersize": 5,
145     "legend.framealpha": 0.92,
146     "legend.edgecolor": "0.75",
147     "savefig.dpi": 300,
148     "savefig.bbox": "tight",
149     "savefig.pad_inches": 0.03,
150 })
151
152
153 def save_figure(fig, folder, name):
154     folder.mkdir(parents=True, exist_ok=True)
155     fig.savefig(folder / f"{name}.pdf")
156     fig.savefig(folder / f"{name}.png")
157     plt.close(fig)
158
159
160 def panel_letter(ax, letter, three_d=False):
161     # letters sit just outside the top-left corner so they never cover data
162     kw = dict(fontsize=10, fontweight="bold")
163     if three_d:
164         ax.text2D(0.0, 0.97, f"({letter})", transform=ax.transAxes, **kw)
165     else:
166         ax.text(-0.02, 1.02, f"({letter})", transform=ax.transAxes,
167                va="bottom", ha="right", **kw)
168
169
170 def field_levels(Y):
171     """Contour levels for f and f_hat maps, plus a colour map that flags undershoot.
172
173     The lower limit sits just below min(f), so round-off never triggers the
174     flag, while a surrogate that predicts clearly impossible values shows up red.
175     """
176     lo, hi = Y.min(), Y.max()
177     cmap = plt.get_cmap("viridis").copy()
178     cmap.set_under("#e34948")
179     return np.linspace(lo - UNDERSHOOT_TOL * (hi - lo), hi, 61), cmap
180
181
182 def latex_sci(v, digits=2):
183     mantissa, exponent = f"{v:.{digits}e}".split("e")
184     return rf"{mantissa}\times 10^{{{int(exponent)}}}"
185
186
187 def sci_colorbar(fig, mappable, ax, label):
188     cb = fig.colorbar(mappable, ax=ax, pad=0.02, fraction=0.05)
189     lo, hi = mappable.norm.vmin, mappable.norm.vmax
190     # keep ticks inside the data range, otherwise extend="both" colour bars
191     # get a tick drawn on the arrow that collides with its neighbour
192     cb.set_ticks([t for t in MaxNLocator(6).tick_values(lo, hi) if lo <= t <= hi])
193     fmt = ScalarFormatter(useMathText=True)
194     fmt.set_powerlimits((-3, 4))
195     cb.formatter = fmt
196     cb.update_ticks()
197     cb.set_label(label)
198     return cb
199
200
201 # ----- design of experiments -----
202
203 def latin_hypercube(n, d, rng):
204     """Classic LHS in the unit cube: one point per bin on every axis."""
205     bins = np.column_stack([rng.permutation(n) for _ in range(d)])
206     return (bins + rng.random((n, d))) / n
207

```

```

208
209 def uniform_random(n, d, rng):
210     return rng.random((n, d))
211
212
213 def maximin_lhs(n, d, rng, n_candidates):
214     """Best of several LHS draws by the maximin criterion.
215
216     Generating a design costs no function evaluations, so screening candidates
217     improves the spread for free while keeping the LHS stratification.
218     """
219     best, best_score = None, -np.inf
220     for _ in range(n_candidates):
221         U = latin_hypercube(n, d, rng)
222         score = pdist(U).min()
223         if score > best_score:
224             best, best_score = U, score
225     return best
226
227
228 def to_box(U, bounds):
229     return bounds[:, 0] + U * (bounds[:, 1] - bounds[:, 0])
230
231
232 def min_distance(U):
233     return pdist(U).min()
234
235
236 def corner_gaps(X_train, bounds):
237     """Distance from each corner of the box to its nearest training point."""
238     corners = np.array([[bounds[0, i], bounds[1, j]] for i in (0, 1) for j in (0, 1)])
239     return {f"({c[0]:g}, {c[1]:g})": float(np.min(np.linalg.norm(X_train - c, axis=1)))
240             for c in corners}
241
242
243 def compare_designs(rng):
244     """Distribution of the maximin criterion over many realisations, in [0,1]^2."""
245     scores = {"Uniform random": [], "LHS": []}
246     for _ in range(DOE_REALISATIONS):
247         scores["Uniform random"].append(min_distance(uniform_random(N_SAMPLES, 2, rng)))
248         scores["LHS"].append(min_distance(latin_hypercube(N_SAMPLES, 2, rng)))
249     return {k: np.array(v) for k, v in scores.items()}
250
251
252 # ----- surrogates -----
253
254 class CubicRBF(RegressorMixin, BaseEstimator):
255     """Cubic RBF interpolant with a linear polynomial tail.
256
257     f_hat(x) = sum_i w_i ||x - x_i||^3 + c0 + c1 x1 + c2 x2, with sum w_i = 0 and
258     sum w_i x_i = 0 so that the weights carry no linear trend of their own.
259     """
260
261     def fit(self, X, y):
262         X, y = np.asarray(X, float), np.asarray(y, float)
263         n, d = X.shape
264         Phi = cdist(X, X) ** 3
265         P = np.hstack([np.ones((n, 1)), X])
266         A = np.block([[Phi, P], [P.T, np.zeros((d + 1, d + 1))]])
267         sol = np.linalg.solve(A, np.concatenate([y, np.zeros(d + 1)]))
268         self.centres_, self.weights_, self.tail_ = X, sol[:n], sol[n:]
269         self.condition_number_ = np.linalg.cond(A)
270         return self
271
272     def predict(self, X):
273         X = np.asarray(X, float)
274         P = np.hstack([np.ones((len(X), 1)), X])
275         return cdist(X, self.centres_) ** 3 @ self.weights_ + P @ self.tail_
276
277
278 def make_polynomial(degree):

```

```

279 # scaling sits inside the pipeline, so every fit (including each CV fold)
280 # learns mu and sigma from its own training rows only - no leakage
281 return make_pipeline(StandardScaler(),
282                      PolynomialFeatures(degree, include_bias=False),
283                      LinearRegression())
284
285
286 def make_rbf():
287     return make_pipeline(StandardScaler(), CubicRBF())
288
289
290 def make_gp(seed=SEED, signal_max=GP_SIGNAL_MAX):
291     kernel = (ConstantKernel(GP_INIT["signal"], (GP_SIGNAL_MIN, signal_max))
292              * RBF(length_scale=GP_INIT["length"], length_scale_bounds=GP_LENGTH_BOUNDS)
293              + WhiteKernel(GP_INIT["noise"], GP_NOISE_BOUNDS))
294     return make_pipeline(StandardScaler(),
295                         GaussianProcessRegressor(kernel=kernel, normalize_y=True, alpha=GP_ALPHA,
296                                                  n_restarts_optimizer=GP_RESTARTS,
297                                                  random_state=seed))
298
299
300 def make_model(name, degree):
301     if name == "Polynomial":
302         return make_polynomial(degree)
303     if name == "Cubic RBF":
304         return make_rbf()
305     return make_gp()
306
307
308 def n_terms(p):
309     return (p + 1) * (p + 2) // 2
310
311
312 def gp_bound_sensitivity(X_train, y_train):
313     """Refit the GP with wider sigma_f^2 bounds, judged by training CV only.
314
315     A wider bound still contains the old optimum, so a lower likelihood after
316     widening means the optimiser failed, not that the model got worse; the
317     abnormal terminations are counted to make that visible.
318     """
319     rows = []
320     for smax in GP_SIGNAL_TRIALS:
321         m = make_gp(signal_max=smax)
322         n_abnormal = fit_gp_counting_failures(m, X_train, y_train)
323         hp = gp_hyperparameters(m)
324         rows.append({"signal_max": smax,
325                    "signal_variance": hp["signal_variance_normalised"],
326                    "ell1_phys": hp["length_scale_phys"][0],
327                    "ell2_phys": hp["length_scale_phys"][1],
328                    "noise_variance": hp["noise_variance_normalised"],
329                    "log_marginal_likelihood": hp["log_marginal_likelihood"],
330                    "CV_RMSE": cv_rmse(make_gp(signal_max=smax), X_train, y_train),
331                    "abnormal_terminations": n_abnormal,
332                    "optimizer_runs": GP_RESTARTS + 1,
333                    "at_bounds": ";\n".join(hp["parameters_at_bounds"])}))
334     return pd.DataFrame(rows)
335
336
337 def gp_hyperparameters(gp_pipeline):
338     """Fitted GP kernel parameters, with length scales in physical units too."""
339     gp = gp_pipeline[-1]
340     scaler = gp_pipeline[0]
341     k = gp.kernel_
342     ell_std = np.atleast_1d(k.k1.k2.length_scale)
343     at_bound = []
344     for hp, theta in zip(k.hyperparameters, k.theta):
345         lo, hi = np.log(hp.bounds[0])
346         if np.isclose(theta, lo, atol=GP_BOUND_TOL) or np.isclose(theta, hi, atol=GP_BOUND_TOL):
347             at_bound.append(hp.name)
348     return {
349         "signal_variance_normalised": float(k.k1.k1.constant_value),

```

```

350     "length_scale_std": ell_std.tolist(),
351     "length_scale_phys": (ell_std * scaler.scale_).tolist(),
352     "noise_variance_normalised": float(k.k2.noise_level),
353     "log_marginal_likelihood": float(gp.log_marginal_likelihood_value_),
354     "parameters_at_bounds": at_bound,
355 }
356
357
358 # ----- validation -----
359
360 def metrics(y_true, y_pred):
361     y_true, y_pred = np.asarray(y_true), np.asarray(y_pred)
362     rmse = float(np.sqrt(mean_squared_error(y_true, y_pred)))
363     y_range = y_true.max() - y_true.min()
364     return {"RMSE": rmse,
365             "MAE": float(mean_absolute_error(y_true, y_pred)),
366             "R2": float(r2_score(y_true, y_pred)),
367             "MaxAE": float(np.max(np.abs(y_true - y_pred))),
368             "NRMSE": rmse / y_range if y_range > 0 else float("nan")}
369
370
371 def cv_rmse(model, X, y):
372     """Pooled out-of-fold RMSE; cross_val_predict refits the whole pipeline per fold."""
373     folds = KFold(CV_FOLDS, shuffle=True, random_state=SEED)
374     with warnings.catch_warnings():
375         # GP bound warnings are checked explicitly on the final model instead
376         warnings.simplefilter("ignore", ConvergenceWarning)
377         pred = cross_val_predict(model, X, y, cv=folds)
378     return float(np.sqrt(mean_squared_error(y, pred)))
379
380
381 def design_matrix_condition(X_train, degree):
382     Z = StandardScaler().fit_transform(X_train)
383     return float(np.linalg.cond(PolynomialFeatures(degree).fit_transform(Z)))
384
385
386 def degree_sweep(X_train, y_train):
387     """Training-only quantities for each degree; nothing here sees the test set."""
388     rows = []
389     fold_size = int(len(X_train) * (CV_FOLDS - 1) / CV_FOLDS)
390     for p in POLY_DEGREES:
391         m = make_polynomial(p).fit(X_train, y_train)
392         rows.append({
393             "degree": p,
394             "n_terms": n_terms(p),
395             "points_per_fold": fold_size,
396             "cond_design": design_matrix_condition(X_train, p),
397             "train_RMSE": metrics(y_train, m.predict(X_train))["RMSE"],
398             "CV_RMSE": cv_rmse(make_polynomial(p), X_train, y_train),
399         })
400     return pd.DataFrame(rows)
401
402
403 def select_degree(sweep, y_train):
404     """Smallest degree whose CV error ties with the best one, or is at round-off.
405
406     With noiseless polynomial data every degree at or above the true one gives
407     round-off error, and the argmin among those is decided by noise. Preferring
408     the simplest adequate degree is the parsimony rule.
409     """
410     threshold = max((1 + PARSIMONY_TOL) * sweep["CV_RMSE"].min(),
411                     ROUNDOFF_REL * np.std(y_train))
412     return int(sweep.loc[sweep["CV_RMSE"] <= threshold, "degree"].min())
413
414
415 def add_sweep_diagnostics(sweep, X_train, y_train, X_test, y_test, X_grid, y_grid):
416     """Test and dense-grid error per degree, added only after the degree is locked."""
417     test, grid = [], []
418     for p in sweep["degree"]:
419         m = make_polynomial(p).fit(X_train, y_train)
420         test.append(metrics(y_test, m.predict(X_test))["RMSE"])

```

```

421     grid.append(metrics(y_grid, m.predict(X_grid))["RMSE"])
422     return sweep.assign(test_RMSE=test, grid_RMSE=grid)
423
424
425 def polynomial_readback(model, degree, X_grid):
426     """Coefficients of the fitted surrogate in the raw monomials of x.
427
428     An affine input scaling keeps the polynomial degree, so f_hat is an exact
429     degree-p polynomial in x and projecting it on raw monomials recovers them.
430     """
431     pf = PolynomialFeatures(degree)
432     Phi = pf.fit_transform(X_grid)
433     coef, *_ = np.linalg.lstsq(Phi, model.predict(X_grid), rcond=None)
434     return dict(zip(pf.get_feature_names_out(["x1", "x2"]), coef))
435
436
437 def quadratic_stationary_point(c):
438     """Solve grad f = 0 for f = c0 + a x1 + b x2 + A x1^2 + B x1 x2 + C x2^2."""
439     H = np.array([[2 * c["x1^2"], c["x1 x2"]], [c["x1 x2"], 2 * c["x2^2"]]])
440     g0 = np.array([c["x1"], c["x2"]])
441     return np.linalg.solve(H, -g0), H
442
443
444 # ----- optimisation -----
445
446 def multistart_minimise(objective, bounds, starts, cluster_tol):
447     """Bounded Nelder-Mead from every start, then merge nearby end points.
448
449     Nelder-Mead is derivative free, which avoids the finite-difference trap on
450     GP predictions noted in tutorial Sec. 15.2.
451     """
452     ends, n_eval = [], 0
453     for x0 in starts:
454         res = minimize(lambda x: float(objective(x[None, :])[0]), x0,
455                        method="Nelder-Mead", bounds=list(map(tuple, bounds)),
456                        options=dict(xatol=NM_XATOL, fatol=NM_FATOL, maxiter=NM_MAXITER))
457         ends.append((float(res.fun), res.x))
458         n_eval += res.nfev
459     ends.sort(key=lambda t: t[0])
460     clusters = []
461     for fval, x in ends:
462         for c in clusters:
463             if np.linalg.norm(x - c["x"]) <= cluster_tol:
464                 c["hits"] += 1
465                 break
466         else:
467             clusters.append({"f_hat": fval, "x": x, "hits": 1})
468     return clusters, n_eval
469
470
471 def on_boundary(x, bounds, tol=BOUNDARY_TOL):
472     return bool(np.any(np.abs(x - bounds[:, 0]) < tol) or np.any(np.abs(x - bounds[:, 1]) < tol))
473
474
475 # ----- figures -----
476
477 def plot_truth(spec, G, out):
478     X1, X2, Y = G["X1"], G["X2"], G["Y"]
479     fig = plt.figure(figsize=(7.2, 3.0))
480     ax = fig.add_subplot(1, 2, 1)
481     cs = ax.contourf(X1, X2, Y, levels=60, cmap="viridis")
482     cs.set_rasterized(True)
483     cl = ax.contour(X1, X2, Y, levels=spec["line_levels"], colors="white",
484                    linewidths=0.6, alpha=0.8)
485     ax.clabel(cl, fontsize=7, fmt="%g")
486     if spec["name"] == "Rosenbrock":
487         x = np.linspace(-np.sqrt(2), np.sqrt(2), 200)
488         ax.plot(x, x**2, "--", color="#eb6834", lw=1.2, label=r"valley $x_2=x_1^2$")
489     else:
490         t = np.linspace(-6, 6, 50)
491         ax.plot(2 + t, 2 + t, "--", color="#eb6834", lw=1.1, label="Hessian eigenvectors")

```

```

492     ax.plot(2 + t, 2 - t, "--", color="#eb6834", lw=1.1)
493     ax.set(xlim=spec["bounds"][0], ylim=spec["bounds"][1])
494     ax.plot(*spec["minima"].T, "*", ms=13, mfc="#e34948", mec="white", mew=0.8,
495           ls="none", label=f"global minimum, $f={spec['f_min']:g}$")
496     ax.set(xlabel="$x_1$", ylabel="$x_2$", aspect="equal")
497     ax.legend(loc="lower right", fontsize=7.5)
498     sci_colorbar(fig, cs, ax, "$f(x_1,x_2)$")
499     panel_letter(ax, "a")
500
501     ax3 = fig.add_subplot(1, 2, 2, projection="3d")
502     ax3.plot_surface(X1, X2, Y, cmap="viridis", linewidth=0, antialiased=True,
503                    rstride=2, cstride=2, rasterized=True)
504     fx, fy = spec["minima"][0]
505     ax3.scatter([fx], [fy], [spec["f_min"]], marker="*", s=90, c="#e34948",
506               edgcolors="white", depthshade=False, zorder=10)
507     ax3.set_xlabel("$x_1$", labelpad=-2)
508     ax3.set_ylabel("$x_2$", labelpad=-2)
509     ax3.set_zlabel("$f$", labelpad=0)
510     ax3.tick_params(pad=-1)
511     ax3.view_init(*spec["view"])
512     ax3.set_box_aspect(None, zoom=0.85)
513     panel_letter(ax3, "b", three_d=True)
514     fig.subplots_adjust(wspace=0.3)
515     save_figure(fig, out, "fig01_truth")
516
517
518 def plot_doe(spec, designs, scores, chosen_score, out):
519     fig, axes = plt.subplots(2, 2, figsize=(6.4, 6.0))
520     b = spec["bounds"]
521     pad = 0.07 * (b[0, 1] - b[0, 0])
522     for ax, (label, U), letter in zip(axes.flat[:3], designs.items(), "abc"):
523         X = to_box(U, b)
524         ax.plot(X[:, 0], X[:, 1], "o", ms=4.2, mfc="#2a78d6", mec="black", mew=0.4, ls="none")
525         # rug ticks along the edges show the one-dimensional projections
526         ax.plot(X[:, 0], np.full(len(X), b[1, 0] - 0.55 * pad), "|", color="#e34948", ms=6)
527         ax.plot(np.full(len(X), b[0, 0] - 0.55 * pad), X[:, 1], "_", color="#e34948", ms=6)
528         ax.set(xlim=(b[0, 0] - pad, b[0, 1] + 0.02 * pad),
529              ylim=(b[1, 0] - pad, b[1, 1] + 0.02 * pad),
530              xlabel="$x_1$", ylabel="$x_2$", aspect="equal")
531         panel_letter(ax, letter)
532     ax = axes[1, 1]
533     bins = np.linspace(0, max(s.max() for s in scores.values()) * 1.05, 40)
534     for (label, s), color in zip(scores.items(), ["#2a78d6", "#eb6834"]):
535         ax.hist(s, bins=bins, histtype="stepfilled", alpha=0.25, color=color)
536         ax.hist(s, bins=bins, histtype="step", lw=1.3, color=color,
537               label=f"{label} (median {np.median(s):.4f})")
538     ax.axvline(chosen_score, color="#4a3aa7", ls="--", lw=1.3, label="maximin LHS used")
539     ax.set(xlabel=r"minimum pairwise distance $\phi_{\min}$ in $[0,1]^2$",
540           ylabel=f"count ({DOE_REALISATIONS} designs)")
541     ax.set_ylim(0, ax.get_ylim()[1] * 1.4) # headroom so the legend clears the bars
542     ax.set_xlim(0, 1.08 * max(chosen_score, bins[-1]))
543     ax.legend(fontsize=7.5, loc="upper left")
544     ax.set_box_aspect(1)
545     panel_letter(ax, "d")
546     fig.tight_layout()
547     save_figure(fig, out, "fig02_doe")
548
549
550 def plot_split(spec, G, X_train, X_test, out):
551     fig, ax = plt.subplots(figsize=(3.9, 3.4))
552     ax.contour(G["X1"], G["X2"], G["Y"], levels=spec["line_levels"], colors="0.55", linewidths=0.6)
553     ax.plot(X_train[:, 0], X_train[:, 1], "o", ms=5, mfc="#2a78d6", mec="black", mew=0.5,
554           ls="none", label=f"training ($n={len(X_train)}$)")
555     ax.plot(X_test[:, 0], X_test[:, 1], "s", ms=5.5, mfc="#eb6834", mec="black", mew=0.5,
556           ls="none", label=f"test ($n={len(X_test)}$)")
557     ax.plot(*spec["minima"].T, "*", ms=13, mfc="#e34948", mec="white", mew=0.8, ls="none",
558           label="global minimum")
559     ax.set(xlabel="$x_1$", ylabel="$x_2$", aspect="equal",
560           xlim=spec["bounds"][0], ylim=spec["bounds"][1])
561     ax.legend(loc="upper center", bbox_to_anchor=(0.5, -0.17), ncol=3, fontsize=7.5,
562           handletextpad=0.3, columnspacing=0.8)

```

```

563 save_figure(fig, out, "fig03_split")
564
565
566 def plot_degree_sweep(sweep, chosen, out):
567     fig, (ax, axc) = plt.subplots(1, 2, figsize=(7.0, 2.8), gridspec_kw={"width_ratios": [1.6, 1]})
568     p = sweep["degree"]
569     under = sweep.loc[sweep["n_terms"] > sweep["points_per_fold"], "degree"]
570     for a in (ax, axc):
571         if len(under):
572             a.axvspan(under.min() - 0.5, p.max() + 0.5, color="0.9", zorder=0)
573             a.axvline(chosen, color="0.35", ls=":", lw=1.2, zorder=1)
574             a.grid(alpha=0.3, lw=0.5)
575             a.set_xticks(list(p))
576     floor = 1e-16
577     series = [
578         ("train_RMSE", "train", "#2a78d6", "o", "-"),
579         ("CV_RMSE", f"{CV_FOLDS}-fold CV (selection)", "#eb6834", "D", "-"),
580         ("test_RMSE", "test (diagnostic)", "#1baf7a", "s", "--"),
581         ("grid_RMSE", "dense grid (diagnostic)", "#4a3aa7", "^", "--")
582     ]
583     for col, label, color, mk, ls in series:
584         ax.semilogy(p, np.maximum(sweep[col], floor), ls=ls, marker=mk, color=color,
585                     ms=4.5, label=label)
586     ax.set(xlabel="polynomial degree $p$", ylabel="RMSE", xlim=(0.5, p.max() + 0.5))
587     ax.legend(fontsize=7.5, loc="lower left")
588     ax.text(chosen + 0.1, 0.97, f"selected $p$={chosen}$", transform=ax.get_xaxis_transform(),
589            va="top", fontsize=7.5, color="0.25")
590     if len(under):
591         ax.text(0.5 * (under.min() + p.max()), 1.01, "terms > points\nper CV fold",
592                transform=ax.get_xaxis_transform(), fontsize=7, color="0.35",
593                ha="center", va="bottom")
594     panel_letter(ax, "a")
595     axc.semilogy(p, sweep["cond_design"], "o-", color="0.15", ms=4.5)
596     axc.set(xlabel="polynomial degree $p$", ylabel=r"cond$(\mathbf{\Psi})$", xlim=(0.5, p.max() + 0.5))
597     panel_letter(axc, "b")
598     fig.tight_layout()
599     save_figure(fig, out, "fig04_degree_sweep")
600
601 def plot_parity(y_test, preds, labels, out):
602     fig, axes = plt.subplots(1, 3, figsize=(7.0, 2.55))
603     lo = min(y_test.min(), min(v.min() for v in preds.values()))
604     hi = max(y_test.max(), max(v.max() for v in preds.values()))
605     pad = 0.05 * (hi - lo)
606     lim = (lo - pad, hi + pad)
607     for ax, name, letter in zip(axes, MODEL_NAMES, "abc"):
608         ax.plot(lim, lim, "-", color="0.5", lw=1.0, zorder=1)
609         ax.plot(y_test, preds[name], MODEL_MARKERS[name], ms=5, mfc=MODEL_COLORS[name],
610                mec="black", mew=0.5, ls="none", zorder=3)
611         m = metrics(y_test, preds[name])
612         ax.text(0.97, 0.04, f"{labels[name]}\nRMSE = {latex_sci(m['RMSE'])}\n$R^2$={m['R2']:.6f}$",
613                transform=ax.transAxes, ha="right", va="bottom", fontsize=7.5,
614                bbox=dict(boxstyle="round,pad=0.25", fc="white", ec="0.7", alpha=0.9))
615         ax.set(xlim=lim, ylim=lim, aspect="equal", xlabel="true $f$ (test set)")
616         ax.grid(alpha=0.3, lw=0.5)
617         panel_letter(ax, letter)
618     axes[0].set_ylabel(r"predicted $\hat{f}$")
619     fig.tight_layout()
620     save_figure(fig, out, "fig05_parity")
621
622 def plot_side_by_side(spec, G, Y_hat, X_train, label, out, name):
623     X1, X2, Y = G["X1"], G["X2"], G["Y"]
624     E = Y - Y_hat
625     levels, cmap = field_levels(Y)
626     emax = np.abs(E).max()
627     fig, axes = plt.subplots(1, 3, figsize=(7.4, 2.35))
628     panels = [(Y, "True $f$", "$f$"), (Y_hat, f"${label}$\hat{f}$", "$\hat{f}$")]
629     for ax, (Z, title, cb_label) in zip(axes[:2], panels):
630         # both panels use the same levels, so a colour means the same value;
631         # extend="both" shows (rather than hides) surrogate values outside them
632         cs = ax.contourf(X1, X2, Z, levels=levels, cmap=cmap, extend="both")
633         cs.set_rasterized(True)

```

```

634     ax.contour(X1, X2, Z, levels=spec["line_levels"], colors="white", linewidths=0.5, alpha=0.75)
635     ax.set_title(title, fontsize=8.5, pad=3)
636     sci_colorbar(fig, cs, ax, cb_label)
637     axes[0].plot(*spec["minima"].T, "*", ms=10, mfc="#e34948", mec="white", mew=0.7, ls="none")
638     axes[1].plot(X_train[:, 0], X_train[:, 1], "o", ms=2.8, mfc="white", mec="black", mew=0.4, ls="none")
639     cs = axes[2].contourf(X1, X2, E, levels=np.linspace(-emax, emax, 41), cmap="RdBu_r",
640                          vmin=-emax, vmax=emax)
641     cs.set_rasterized(True)
642     axes[2].plot(X_train[:, 0], X_train[:, 1], "o", ms=3, mfc="#eda100", mec="black", mew=0.45,
643                ls="none")
644     axes[2].set_title("Error  $f - \hat{f}$ ", fontsize=8.5, pad=3)
645     sci_colorbar(fig, cs, axes[2], "f -  $\hat{f}$ ")
646     for ax in axes:
647         ax.set(xlabel="$x_1$", ylabel="$x_2$", aspect="equal")
648     fig.tight_layout(w_pad=0.6)
649     save_figure(fig, out, f"fig06_side_by_side_{name}")
650
651
652 def plot_surfaces(spec, G, Y_hat, X_train, y_train, label, out, name):
653     X1, X2, Y = G["X1"], G["X2"], G["Y"]
654     vmin, vmax = Y.min(), Y.max()
655     zlim = (min(vmin, Y_hat.min()), max(vmax, Y_hat.max()))
656     fig = plt.figure(figsize=(7.0, 3.0))
657     for i, (Z, title) in enumerate([(Y, "True  $f$ "), (Y_hat, f"{label}  $\hat{f}$ )]):
658         ax = fig.add_subplot(1, 2, i + 1, projection="3d")
659         surf = ax.plot_surface(X1, X2, Z, cmap="viridis", vmin=vmin, vmax=vmax, linewidth=0,
660                              rstride=2, cstride=2, antialiased=True, rasterized=True)
661         if i == 1:
662             ax.scatter(X_train[:, 0], X_train[:, 1], y_train, c="#e34948", s=7,
663                      depthshade=False, label="training data")
664             ax.legend(loc="upper right", fontsize=7.5)
665             ax.set(xlabel="$x_1$", ylabel="$x_2$", zlabel="$f$ if  $i == 0$  else  $r \hat{f}$ ",
666                  zlim=zlim, xlim=spec["bounds"][0], ylim=spec["bounds"][1])
667             ax.set_title(title, fontsize=8.5, pad=0)
668             ax.view_init(*spec["view"])
669             ax.set_box_aspect(None, zoom=0.86)
670             fig.colorbar(surf, ax=fig.axes, shrink=0.7, pad=0.04, label="$f$,  $\hat{f}$  (shared scale)")
671             save_figure(fig, out, f"fig07_surfaces_{name}")
672
673
674 def plot_uncertainty(G, sigma, abs_err, X_train, out):
675     fig = plt.figure(figsize=(7.4, 2.45))
676     axes = [fig.add_subplot(1, 3, i + 1) for i in range(3)]
677     # both fields span several decades and peak in one corner, so a linear
678     # scale would show a single colour; the log scale keeps the interior visible
679     for ax, Z, label, letter in [(axes[0], sigma, r"GP std. dev.  $\hat{\sigma}$ ", "a"),
680                                 (axes[1], abs_err, r"true  $|f - \hat{f}|$ ", "b")]:
681         Zc = np.maximum(Z, Z.max() * 1e-5)
682         levels = np.logspace(np.log10(Zc.min()), np.log10(Zc.max()), 41)
683         cs = ax.contourf(G["X1"], G["X2"], Z, levels=levels, cmap="cividis", norm=LogNorm())
684         cs.set_rasterized(True)
685         ax.plot(X_train[:, 0], X_train[:, 1], "o", ms=2.8, mfc="white", mec="black", mew=0.4, ls="none")
686         ax.set(xlabel="$x_1$", ylabel="$x_2$", aspect="equal")
687         cb = fig.colorbar(cs, ax=ax, pad=0.02, fraction=0.05, label=label)
688         cb.locator = LogLocator()
689         cb.update_ticks()
690         panel_letter(ax, letter)
691     ax = axes[2]
692     s, e = sigma.ravel(), abs_err.ravel()
693     keep = np.random.default_rng(SEED).choice(len(s), SCATTER_POINTS, replace=False)
694     ax.loglog(s[keep], np.maximum(e[keep], 1e-16), ".", ms=2, color="#1baf7a", alpha=0.5)
695     ref = np.logspace(np.log10(s.min()), np.log10(s.max()), 10)
696     ax.loglog(ref, 2 * ref, "---", color="0.2", lw=1.0, label=r"$|e| = 2 \hat{\sigma}$")
697     ax.set(xlabel=r"$\hat{\sigma}$", ylabel=r"$|f - \hat{f}|$")
698     ax.legend(fontsize=7.5, loc="upper left")
699     ax.set_box_aspect(1)
700     ax.grid(alpha=0.3, lw=0.5, which="major")
701     panel_letter(ax, "c")
702     fig.tight_layout(w_pad=0.5)
703     save_figure(fig, out, "fig08_gp_uncertainty")
704

```

```

705
706 def plot_learning_curve(curve, y_grid_std, out):
707     fig, axes = plt.subplots(1, 2, figsize=(7.0, 2.9), sharey=True)
708     for ax, col, ylabel, letter in [(axes[0], "grid_RMSE", "RMSE on the dense grid", "a"),
709                                     (axes[1], "core_RMSE", "", "b")]:
710         for name in MODEL_NAMES:
711             d = curve[curve["model"] == name].groupby("N")[col]
712             med, lo, hi = d.median(), d.min(), d.max()
713             n = med.index.values
714             # band = spread over the repeated LHS draws, line = median
715             ax.fill_between(n, np.maximum(lo, 1e-15), np.maximum(hi, 1e-15),
716                             color=MODEL_COLORS[name], alpha=0.15, lw=0)
717             ax.loglog(n, np.maximum(med, 1e-15), marker=MODEL_MARKERS[name],
718                       color=MODEL_COLORS[name], ms=4.5, label=name)
719             ax.axhline(y_grid_std, color="0.4", ls=":", lw=1.2, label="mean predictor (std of $f$)")
720             ax.set_xlabel("training points $N$", ylabel=ylabel)
721             ax.set_xticks(LEARNING_N)
722             ax.set_xticklabels([str(n) for n in LEARNING_N])
723             ax.xaxis.set_minor_locator(plt.NullLocator())
724             ax.grid(alpha=0.3, lw=0.5)
725             panel_letter(ax, letter)
726         axes[1].set_ylabel(f"RMSE in the central box ({int(CORE_FRAC * 100)}% of each side)")
727         axes[1].yaxis.set_tick_params(labelleft=True)
728         handles, names = axes[0].get_legend_handles_labels()
729         fig.tight_layout(rect=(0, 0.08, 1, 1))
730         fig.legend(handles, names, loc="lower center", ncol=4, fontsize=7.5)
731         save_figure(fig, out, "fig09_learning_curve")
732
733
734 def plot_optimisation(spec, G, surfaces, found, starts, out):
735     levels, cmap = field_levels(G["Y"])
736     fig, axes = plt.subplots(1, 3, figsize=(7.4, 2.55))
737     for ax, name, letter in zip(axes, MODEL_NAMES, "abc"):
738         cs = ax.contourf(G["X1"], G["X2"], surfaces[name], levels=levels, cmap=cmap,
739                         extend="both")
740         cs.set_rasterized(True)
741         ax.contour(G["X1"], G["X2"], surfaces[name], levels=spec["line_levels"], colors="white",
742                   linewidths=0.45, alpha=0.7)
743         ax.plot(starts[:, 0], starts[:, 1], ".", ms=3, color="white", alpha=0.9)
744         xs = np.array([c["x"] for c in found[name]])
745         ax.plot(*spec["minima"].T, "*", ms=13, mfc="#e34948", mec="white", mew=0.8, ls="none")
746         ax.plot(xs[:, 0], xs[:, 1], "X", ms=8, mfc="#eda100", mec="black", mew=0.7, ls="none")
747         ax.set_xlabel("$x_1$", ylabel="$x_2$" if letter == "a" else "", aspect="equal")
748         ax.text(0.03, 0.04, transform=ax.transAxes, fontsize=7.5, color="white")
749         panel_letter(ax, letter)
750         handles = [plt.Line2D([], [], marker=".", color="white", mec="0.3", ls="none", ms=6),
751                   plt.Line2D([], [], marker="*", mfc="#e34948", mec="white", ls="none", ms=11),
752                   plt.Line2D([], [], marker="X", mfc="#eda100", mec="black", ls="none", ms=8)]
753         fig.legend(handles, ["start points", "known global minimum", "surrogate minima"],
754                   loc="lower center", ncol=3, fontsize=7.5, bbox_to_anchor=(0.45, -0.07))
755         cb = fig.colorbar(cs, ax=axes, pad=0.015, fraction=0.03, label=r"$\hat{f}$")
756         cb.set_ticks([t for t in MaxNLocator(6).tick_values(levels[0], levels[-1])
757                      if levels[0] <= t <= levels[-1]])
758         save_figure(fig, out, "fig10_optimisation")
759
760
761 def plot_extrapolation(spec, G, Y_hat, X_sub, label, out, name):
762     X1, X2, Y = G["X1"], G["X2"], G["Y"]
763     levels, cmap = field_levels(Y)
764     E = Y - Y_hat
765     emax = np.abs(E).max()
766     sb = spec["sub_box"]
767     fig, axes = plt.subplots(1, 3, figsize=(7.4, 2.35))
768     for ax, Z, title, cb_label in [(axes[0], Y, "True $f$", "$f$"),
769                                    (axes[1], Y_hat, f"{label} $\hat{f}$ (sub-box training)",
770                                     "$\hat{f}$"),
771                                    (axes[2], X_sub, f"{label} $\hat{f}$ (sub-box extrapolation)",
772                                     "$\hat{f}$")]:
773         # outside the sub-box the surrogate may leave the true range; extend="both"
774         # shows that instead of rescaling the colours to hide it
775         cs = ax.contourf(X1, X2, Z, levels=levels, cmap=cmap, extend="both")
776         cs.set_rasterized(True)
777         ax.set_title(title, fontsize=8.5, pad=3)

```

```

776     sci_colorbar(fig, cs, ax, cb_label)
777     axes[1].plot(X_sub[:, 0], X_sub[:, 1], "o", ms=2.6, mfc="white", mec="black", mew=0.4, ls="none")
778     cs = axes[2].contourf(X1, X2, E, levels=np.linspace(-emax, emax, 41), cmap="RdBu_r")
779     cs.set_rasterized(True)
780     axes[2].set_title("Error  $f - \hat{f}$ ", fontsize=8.5, pad=3)
781     sci_colorbar(fig, cs, axes[2], " $f - \hat{f}$ ")
782     for ax in axes:
783         ax.add_patch(Rectangle((sb[0, 0], sb[1, 0]), sb[0, 1] - sb[0, 0], sb[1, 1] - sb[1, 0],
784                                fill=False, ec="#e34948", lw=1.6))
785         ax.set(xlabel=" $x_1$ ", ylabel=" $x_2$ ", aspect="equal")
786     fig.tight_layout(w_pad=0.6)
787     save_figure(fig, out, f"fig11_extrapolation_{name}")
788
789 # ----- the study -----
790
791 def fit_gp_counting_failures(model, X, y):
792     """Fit a GP pipeline and count optimiser runs that ended abnormally.
793
794     The warnings are recorded rather than silenced, so the report can state
795     how often the marginal-likelihood search failed to converge cleanly.
796     """
797     with warnings.catch_warnings(record=True) as caught:
798         warnings.simplefilter("always", ConvergenceWarning)
799         model.fit(X, y)
800     messages = [str(w.message) for w in caught if isinstance(w.category, ConvergenceWarning)]
801     return sum("ABNORMAL" in msg for msg in messages)
802
803
804 def run_study(key, spec):
805     print(f"\n{spec['name']} function")
806     f = spec["func"]
807     b = spec["bounds"]
808     fig_dir = OUTPUT_DIR / "figures" / key
809     tab_dir = OUTPUT_DIR / "tables" / key
810     tab_dir.mkdir(parents=True, exist_ok=True)
811     summary = {"function": spec["name"], "bounds": b.tolist()}
812
813     def f_vec(X):
814         return f(X[:, 0], X[:, 1])
815
816     # step 1: define the truth and check it at the known minimum before fitting
817     x1 = np.linspace(*b[0], GRID_N)
818     x2 = np.linspace(*b[1], GRID_N)
819     X1, X2 = np.meshgrid(x1, x2)
820     G = {"X1": X1, "X2": X2, "Y": f(X1, X2)}
821     X_grid = np.column_stack([X1.ravel(), X2.ravel()])
822     y_grid = G["Y"].ravel()
823     summary["truth"] = {"f_at_known_min": float(f(*spec["minima"][0])),
824                        "grid_min": float(y_grid.min()), "grid_max": float(y_grid.max()),
825                        "grid_mean": float(y_grid.mean()), "grid_std": float(y_grid.std())}
826     plot_truth(spec, G, fig_dir)
827
828     # step 2: designs compared in the unit square with the same budget
829     rng = np.random.default_rng(SEED)
830     designs = {"Uniform random": uniform_random(N_SAMPLES, 2, rng),
831               "LHS": latin_hypcube(N_SAMPLES, 2, rng),
832               "Maximin LHS": maximin_lhs(N_SAMPLES, 2, rng, MAXIMIN_CANDIDATES)}
833     scores = compare_designs(np.random.default_rng(SEED_DOE_STUDY))
834     summary["doe"] = {k: float(min_distance(U)) for k, U in designs.items()}
835     summary["doe_realisations"] = {k: {"median": float(np.median(s)), "mean": float(s.mean()),
836                                       "p05": float(np.percentile(s, 5)), "max": float(s.max())}
837                                   for k, s in scores.items()}
838     plot_doe(spec, designs, scores, summary["doe"]["Maximin LHS"], fig_dir)
839
840     X_doe = to_box(designs["Maximin LHS"], b)
841     y_doe = f_vec(X_doe) # the N expensive evaluations
842
843     # step 3: the test rows are set aside here and only used after all choices are fixed
844     X_train, X_test, y_train, y_test = train_test_split(
845         X_doe, y_doe, test_size=TEST_FRACTION, random_state=SEED)

```

```

847 split = pd.concat([pd.DataFrame({"set": "train", "x1": X_train[:, 0], "x2": X_train[:, 1], "f":
    ↳ y_train}),
848                      pd.DataFrame({"set": "test", "x1": X_test[:, 0], "x2": X_test[:, 1], "f": y_test})
    ↳ ]))
849 split["dist_to_x_min"] = np.min(np.linalg.norm(
850     split[["x1", "x2"]].to_numpy()[ :, None, :] - spec["minima"][None, :, :], axis=2), axis=1)
851 split.to_csv(tab_dir / "doe_split.csv", index=False)
852 summary["split"] = {"n_train": len(X_train), "n_test": len(X_test),
853                    "y_test_range": float(y_test.max() - y_test.min()),
854                    "corner_gaps": corner_gaps(X_train, b)}
855 plot_split(spec, G, X_train, X_test, fig_dir)
856
857 # steps 4-5: polynomial degree chosen by training-only CV, then locked;
858 # the test and grid curves are attached afterwards purely as diagnostics
859 sweep = degree_sweep(X_train, y_train)
860 degree = select_degree(sweep, y_train)
861 sweep = add_sweep_diagnostics(sweep, X_train, y_train, X_test, y_test, X_grid, y_grid)
862 sweep.to_csv(tab_dir / "degree_sweep.csv", index=False)
863 summary["selected_degree"] = degree
864 plot_degree_sweep(sweep, degree, fig_dir)
865
866 models = {name: make_model(name, degree) for name in MODEL_NAMES}
867 models["Polynomial"].fit(X_train, y_train)
868 models["Cubic RBF"].fit(X_train, y_train)
869 n_abnormal = fit_gp_counting_failures(models["Gaussian process"], X_train, y_train)
870 labels = {"Polynomial": f"Polynomial ($p={degree}$)", "Cubic RBF": "Cubic RBF",
871          "Gaussian process": "GP"}
872
873 readback = polynomial_readback(models["Polynomial"], degree, X_grid)
874 exact = spec["exact_coefs"]
875 coef_rows = [{"term": t, "fitted": c, "exact": exact.get(t, 0.0),
876              "abs_diff": abs(c - exact.get(t, 0.0))} for t, c in readback.items()]
877 pd.DataFrame(coef_rows).to_csv(tab_dir / "polynomial_coefficients.csv", index=False)
878 summary["coef_max_abs_diff"] = float(max(r["abs_diff"] for r in coef_rows))
879 summary["rbf_condition_number"] = float(models["Cubic RBF"][-1].condition_number_)
880 summary["gp"] = gp_hyperparameters(models["Gaussian process"])
881 summary["gp"]["optimizer_runs"] = GP_RESTARTS + 1
882 summary["gp"]["abnormal_terminations"] = n_abnormal
883 if summary["gp"]["parameters_at_bounds"]:
884     gp_bound_sensitivity(X_train, y_train).to_csv(tab_dir / "gp_bound_sensitivity.csv",
885                                                  index=False)
886
887 if key == "trid":
888     # hand derivation: grad f = 0 -> [2 -1; -1 2] x = [2; 2]
889     H_true = np.array([[2.0, -1.0], [-1.0, 2.0]])
890     x_hand = np.linalg.solve(H_true, np.array([2.0, 2.0]))
891     x_fit, H_fit = quadratic_stationary_point(readback)
892     summary["hand_check"] = {"x_hand": x_hand.tolist(), "f_hand": float(f(*x_hand)),
893                            "hessian_eigenvalues": np.linalg.eigvalsh(H_true).tolist(),
894                            "x_from_fitted_quadratic": x_fit.tolist(),
895                            "fitted_hessian": H_fit.tolist()}
896
897 # step 6: held-out metrics against the mean-of-training baseline
898 preds = {name: m.predict(X_test) for name, m in models.items()}
899 rows = [{"model": "Baseline mean(y_train)",
900         **metrics(y_test, np.full_like(y_test, y_train.mean()))}]
901 rows += [{"model": labels[n].replace("$", ""), **metrics(y_test, preds[n])} for n in MODEL_NAMES]
902 test_table = pd.DataFrame(rows)
903 test_table.to_csv(tab_dir / "test_metrics.csv", index=False)
904 plot_parity(y_test, preds, labels, fig_dir)
905
906 # step 7: CV on the training partition, with a dense-grid diagnostic next to it
907 grid_pred = {n: models[n].predict(X_grid) for n in ("Polynomial", "Cubic RBF")}
908 grid_pred["Gaussian process"], gp_std = models["Gaussian process"].predict(X_grid, return_std=True)
909 cv_rows = [{"model": labels[n].replace("$", ""),
910            "CV_RMSE": cv_rmse(make_model(n, degree), X_train, y_train),
911            "test_RMSE": metrics(y_test, preds[n])["RMSE"],
912            **{f"grid_{k}": v for k, v in metrics(y_grid, grid_pred[n]).items()}]
913            for n in MODEL_NAMES]
914 cv_table = pd.DataFrame(cv_rows)
915 cv_table.to_csv(tab_dir / "cv_and_grid.csv", index=False)

```

```

916 summary["best_by_cv"] = MODEL_NAMES[int(np.argmax(cv_table["CV_RMSE"]))]
917
918 # step 8: side-by-side maps for every model. The CV-best polynomial is the
919 # required figure, but its error map is pure round-off here, so the RBF and
920 # GP maps are the ones that show where a model without the true form fails
921 surfaces = {n: grid_pred[n].reshape(X1.shape) for n in MODEL_NAMES}
922 tags = {"Polynomial": "poly", "Cubic RBF": "rbf", "Gaussian process": "gp"}
923 for n in MODEL_NAMES:
924     plot_side_by_side(spec, G, surfaces[n], X_train, labels[n], fig_dir, tags[n])
925     plot_surfaces(spec, G, surfaces[n], X_train, y_train, labels[n], fig_dir, tags[n])
926
927 # the core region drops the outer band, to separate edge effects from the
928 # interior; it is used for the uncertainty floor and the learning curve
929 centre, half = b.mean(axis=1), 0.5 * CORE_FRAC * (b[:, 1] - b[:, 0])
930 core = np.all(np.abs(X_grid - centre) <= half, axis=1)
931
932 # step 9: GP standard deviation against the error it cannot see
933 abs_err = np.abs(G["Y"] - surfaces["Gaussian process"])
934 sigma = gp_std.reshape(X1.shape)
935 outside = abs_err > 2 * sigma
936 # jitter plus noise put a floor under sigma_hat; where sigma_hat sits on it,
937 # it carries no information about where the error is
938 gp = models["Gaussian process"][-1]
939 sigma_floor = np.sqrt(gp.alpha + gp.kernel_.k2.noise_level) * y_train.std()
940 summary["gp_uncertainty"] = {
941     "pearson": float(np.corrcoef(sigma.ravel(), abs_err.ravel())[0, 1]),
942     "spearman": float(spearmanr(sigma.ravel(), abs_err.ravel())[0]),
943     "coverage_2sigma": float(np.mean(~outside)),
944     "sigma_max": float(sigma.max()), "abs_err_max": float(abs_err.max()),
945     # where the 2-sigma band fails matters more than how often
946     "abs_err_max_outside_band": float(abs_err[outside].max()) if outside.any() else 0.0,
947     "centroid_outside_band": ([float(X1[outside].mean()), float(X2[outside].mean())]
948                               if outside.any() else None),
949     "sigma_floor": float(sigma_floor),
950     "sigma_at_train_max": float(models["Gaussian process"].predict(
951         X_train, return_std=True)[1].max()),
952     "sigma_median_core": float(np.median(gp_std[core]))}
953 plot_uncertainty(G, sigma, abs_err, X_train, fig_dir)
954
955 # step 10: learning curve on fixed grid, several LHS draws per size
956 lc_rows = []
957 for n in LEARNING_N:
958     p = degree
959     while n_terms(p) > n and p > 1: # keep the least-squares problem determined
960         p -= 1
961     for r in range(LEARNING_REPEATS):
962         U = latin_hypcube(n, 2, np.random.default_rng(SEED + SEED_STRIDE * r + n))
963         Xn = to_box(U, b)
964         yn = f_vec(Xn)
965         for name in MODEL_NAMES:
966             with warnings.catch_warnings():
967                 warnings.simplefilter("ignore", ConvergenceWarning)
968                 m = make_model(name, p).fit(Xn, yn)
969                 yg = m.predict(X_grid)
970                 lc_rows.append({"N": n, "repeat": r, "model": name, "degree_used": p,
971                               "grid_RMSE": metrics(y_grid, yg)["RMSE"],
972                               "core_RMSE": metrics(y_grid[core], yg[core])["RMSE"],
973                               "rbf_condition_number": (m[-1].condition_number_
974                                                         if name == "Cubic RBF" else np.nan)})
975 curve = pd.DataFrame(lc_rows)
976 curve.to_csv(tab_dir / "learning_curve.csv", index=False)
977 lc_summary = (curve.groupby(["model", "N"])[["grid_RMSE", "core_RMSE", "rbf_condition_number"]]
978              .agg(["median", "min", "max"]))
979 lc_summary.columns = ["_".join(c) for c in lc_summary.columns]
980 lc_summary.reset_index().to_csv(tab_dir / "learning_curve_summary.csv", index=False)
981 summary["core_area_fraction"] = CORE_FRAC ** 2
982 plot_learning_curve(curve, y_grid.std(), fig_dir)
983
984 # step 11: optimise every surrogate, then pay one true evaluation per candidate
985 cluster_tol = CLUSTER_FRAC * (b[0, 1] - b[0, 0])
986 starts = to_box(latin_hypcube(N_STARTS, 2, np.random.default_rng(SEED_STARTS)), b)

```

```

987 found, opt_rows, n_verify = {}, [], 0
988 for name in MODEL_NAMES:
989     clusters, _ = multistart_minimise(models[name].predict, b, starts, cluster_tol)
990     found[name] = clusters
991     for rank, c in enumerate(clusters, start=1):
992         f_true = float(f(*c["x"]))
993         n_verify += 1
994         opt_rows.append({
995             "model": name, "rank": rank, "hits": c["hits"],
996             "x1": c["x"][0], "x2": c["x"][1],
997             "f_hat": c["f_hat"], "f_true": f_true,
998             "abs_pred_error": abs(c["f_hat"] - f_true),
999             "gap_to_f_min": f_true - spec["f_min"],
1000             "dist_to_x_min": float(np.min(np.linalg.norm(spec["minima"] - c["x"], axis=1))),
1001             "on_boundary": on_boundary(c["x"], b)}
1002 true_clusters, true_nfev = multistart_minimise(f_vec, b, starts, cluster_tol)
1003 for rank, c in enumerate(true_clusters, start=1):
1004     opt_rows.append({
1005         "model": "True function (check)", "rank": rank, "hits": c["hits"],
1006         "x1": c["x"][0], "x2": c["x"][1], "f_hat": np.nan, "f_true": c["f_hat"],
1007         "abs_pred_error": np.nan, "gap_to_f_min": c["f_hat"] - spec["f_min"],
1008         "dist_to_x_min": float(np.min(np.linalg.norm(spec["minima"] - c["x"], axis=1))),
1009         "on_boundary": on_boundary(c["x"], b)}
1010 opt_table = pd.DataFrame(opt_rows)
1011 opt_table.to_csv(tab_dir / "optimisation.csv", index=False)
1012 summary["optimisation"] = {"n_starts": N_STARTS, "cluster_tol": cluster_tol,
1013                           "true_function_nfev": int(true_nfev)}
1014 plot_optimisation(spec, G, surfaces, found, starts, fig_dir)
1015
1016 # step 12: train on a reduced box, then look inside and outside it separately
1017 sb = spec["sub_box"]
1018 X_sub = to_box(latin_hypercube(len(X_train), 2, np.random.default_rng(SEED_SUB_BOX)), sb)
1019 y_sub = f_vec(X_sub)
1020 inside = np.all((X_grid >= sb[:, 0]) & (X_grid <= sb[:, 1]), axis=1)
1021 ex_rows = []
1022 for name in MODEL_NAMES:
1023     with warnings.catch_warnings():
1024         warnings.simplefilter("ignore", ConvergenceWarning)
1025         m = make_model(name, degree).fit(X_sub, y_sub)
1026         yg = m.predict(X_grid)
1027         m_in, m_out = metrics(y_grid[inside], yg[inside]), metrics(y_grid[~inside], yg[~inside])
1028         ex_rows.append({"model": name, "RMSE_inside": m_in["RMSE"], "RMSE_outside": m_out["RMSE"],
1029                        "MaxAE_inside": m_in["MaxAE"], "MaxAE_outside": m_out["MaxAE"],
1030                        "RMSE_ratio_out_in": m_out["RMSE"] / m_in["RMSE"]})
1031     if name != "Polynomial":
1032         plot_extrapolation(spec, G, yg.reshape(X1.shape), X_sub, labels[name], fig_dir, tags[name])
1033 pd.DataFrame(ex_rows).to_csv(tab_dir / "extrapolation.csv", index=False)
1034
1035 # budget kept apart from the extra evaluations used only for teaching checks
1036 summary["evaluations"] = {
1037     "main_doe": N_SAMPLES,
1038     "dense_grid": GRID_N**2,
1039     "optimum_verification": n_verify,
1040     "true_function_multistart": int(true_nfev),
1041     "learning_curve": int(sum(LEARNING_N) * LEARNING_REPEATS),
1042     "reduced_box": len(X_sub)}
1043
1044 # a short console summary; the full tables are in the CSV files
1045 print(f" selected polynomial degree : {degree} (training-only {CV_FOLDS}-fold CV)")
1046 for _, row in test_table.iterrows():
1047     print(f" test {row['model'][:24s]: RMSE = {row['RMSE']:.3e}, NRMSE = {row['NRMSE']:.3e}")
1048 for name in MODEL_NAMES:
1049     best = found[name][0]
1050     print(f" {name[:18s]} optimum at ({best['x'][0]:.4f}, {best['x'][1]:.4f}), "
1051           f"true f = {f(*best['x']):.6g}")
1052 return summary
1053
1054
1055 def environment_info():
1056     return {"python": platform.python_version(), "numpy": np.__version__, "scipy": scipy.__version__,
1057           "scikit-learn": sklearn.__version__, "matplotlib": matplotlib.__version__,

```

```
1058         "pandas": pd.__version__, "platform": platform.platform(), "seed": SEED}
1059
1060
1061 def main():
1062     set_plot_style()
1063     OUTPUT_DIR.mkdir(parents=True, exist_ok=True)
1064     env = environment_info()
1065     print("Environment:", env)
1066     results = {"environment": env}
1067     for key, spec in FUNCTIONS.items():
1068         results[key] = run_study(key, spec)
1069     with open(OUTPUT_DIR / "summary.json", "w") as fh:
1070         json.dump(results, fh, indent=2, default=float)
1071     print(f"\nAll figures and tables written to {OUTPUT_DIR}")
1072
1073
1074 if __name__ == "__main__":
1075     main()
```